

Lecture 1: Course Introduction

2025

Xin Li
School of Computer Science,
Beijing Institute of Technology

Outline

- Introduction to ML
- Course website and logistics
- Passwd **@lixin**



← → ↻ 🌐 lexue.bit.edu.cn/course/view.php?id=16358#section-0



乐学

机器学习初步 (25春)

网络教室 / 我的课程 / 2024-2025第二学期本科生 / 计算机公共课 / 机器学习初步 (25春)

✚ 📄 新闻通告 ✎

✚ 授课教师: 礼欣

个人主页: <https://cs.bit.edu.cn/szdw/jsml/bssds/e0a7ea1ec73740e2ac05891eede79a10.htm>

BIT1029强化学习小组主页: <https://bit1029public.github.io/>

助教: To be announced

授课地点: 良乡校区文萃楼F404 (周一)、文萃楼I402 (周五)

授课时间: 6-7节 (周一、周五)

标记完成



姓名: 礼欣

所在学科: 计算机科学与技术

职称: 副教授 博士生导师

联系电话: (86) 139-1140-chr(54)+chr(51)+chr(48)+chr(55)

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○ 个人信息

礼欣, 博士, 博士后。

吉林大学学士、硕士, 师从孙吉贵教授。香港浸会大学(Hong Kong Baptist University)博士(导师: William K. Cheung教授与 Jiming Liu教授)、博士后, University of Waterloo 访问学者(Host: Prof. Pascal Poupart), University of Technology Sydney 高级访问学者(Host: Prof. Ivor W. Tsang)。目前主要从事强化学习、图学习的理论研究和技术应用。近年来以第一作者或通讯作者在ICML、NeurIPS、ICLR、KDD、IJCAI、AAAI、TPAMI、TCYB、TKDE、TOIS等人工智能、机器学习领域的知名国际会议及期刊发表多篇学术论文。2010-2019年曾担任The IEEE Intelligent Informatics Bulletin助理执行编辑。

在研究生培养方面, 重视学生的国际交流, 积极推荐学生赴香港浸会大学、香港理工大学、斯坦福大学、悉尼科技大学、新加坡A*STAR等世界知名学府及学术研究机构进行交流访问。指导本科生、硕士研究生、博士研究生发表多篇CCF A类会议论文, CCF A类/SCI一区期刊论文, 并荣获国家奖学金。毕业研究生去向包括: 微软、阿里巴巴、字节跳动、腾讯、京东、美团、滴滴等知名企业, 或中国招商银行总行, 中国工商银行总行, 中科院电子所, 中科院信工所, 中国外交部等企事业单位。

○ 科研方向

机器学习、深度(强化)学习、表示学习理论及应用, 包括:

复杂网络数据挖掘、城市计算、医疗及公共卫生领域的应用, 如: 疾病诊断/预测;

以及深度学习/深度强化学习技术在工业大数据、机器人及特定领域的应用。

○ 代表性学术成果

以下文章列表按发表时年倒序排列(*为通讯作者)

[1] Qianyu Chen, **Xin Li***, Yujie Fang, Mingzhong Wang: Advancing Confidence Calibration and Quantification in Medication Recommendation. KDD 2025, Canada (Accepted)(CCF A)

[2] Ruixiang Sun, Hongyu Zang, **Xin Li***, Riashat Islam: Learning Latent Dynamic Robust Representations for World Models. ICML 2024, Austria (CCF A)

[3] Yujie Fang, **Xin Li***, et al.: Improving GNN Calibration with Discriminative Ability: Insights and Strategies. AAAI Conference on Artificial Intelligence (AAAI), February, 2024, Canada. (CCF A)

[4] Min Wang, **Xin Li***, et al.: MetaCARD: Meta-Reinforcement Learning with Task Uncertainty Feedback via Decoupled Context-Aware Reward and Dynamics Components, AAAI Conference on Artificial Intelligence (AAAI), February, 2024, Canada. (CCF A)

[5] Hongyu Zang, **Xin Li***, et al.: Understanding and Addressing the Pitfalls of Bisimulation-based Representations in Offline Reinforcement Learning. Thirty-seventh Conference on Neural Information Processing Systems 2023 (NeurIPS 2023). (CCF A)

[6] **Xin Li***, et al.: Differentiable Logic Policy for Interpretable Deep Reinforcement Learning: A Study From an Optimization Perspective. IEEE Trans. Pattern Anal. Mach. Intell. 45(10): 11654-11667 (2023) (CCF A)

[7] Hongyu Zang, **Xin Li***, et al.: Behavior Prior Representation learning for Offline Reinforcement Learning. Eleventh International Conference on Learning Representations, ICLR 2023

[8] Fuhao Yang, **Xin Li***, et al.: WaveForM: Graph Enhanced Wavelet Learning for Long Sequence Forecasting of Multivariate Time Series. AAAI Conference on Artificial Intelligence (AAAI), Pages:10754-10761, February, 2023, virtual, United States.(CCF A)

[9] Qianyu Chen, **Xin Li***, et al.: Context-aware safe medication recommendations with molecular graph and DDI graph embedding. AAAI Conference on Artificial Intelligence (AAAI), pages:7053-7060, February, 2023, virtual, United States. (CCF A)

[10] Riashat Islam, Hongyu Zang, Anirudh Goyal, Alex Lamb, Kenji Kawaguchi, **Xin Li**, Romain Laroche, Yoshua Bengio, Remi Tachet des Combes: Discrete Factorial Representations as an Abstraction for Goal Conditioned Reinforcement Learning. Thirty-sixth Conference on Neural Information Processing Systems 2022 (NeurIPS 2022) (CCF A)

[11] Huiting Hong, **Xin Li***, Yuangang Pan, Ivor W. Tsang: Domain-Adversarial Network Alignment. IEEE Trans. Knowl. Data Eng. 34(7): 3211-3224 (2022) (CCF A)

[12] Hongyu Zang, **Xin Li***, Mingzhong Wang: SimSR: Simple Distance-Based State Representations for Deep Reinforcement Learning. AAAI Conference on Artificial Intelligence (AAAI), Pages:8997-9005, February, 2022, virtual, Canada. (CCF A)

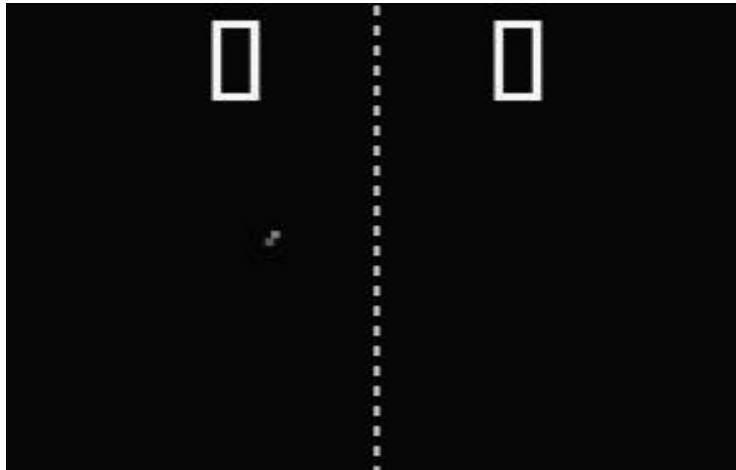
[13] Li Zhang, **Xin Li***, et al.:Off-Policy Differentiable Logic Reinforcement Learning. ECML/PKDD (2) 2021: 617-632, Sept.2021, Basque Country, Spain. (CCF B)

[14] Li Liu, **Xin Li***, William K. Cheung, Lejian Liao: Structural Representation Learning for User Alignment Across Social Networks. IEEE Trans. Knowl. Data Eng. 32(9): 1824-1837 (2020) (CCF A)

[15] Li Zhang, **Xin Li***, et al.: Universal Value Iteration Networks: When Spatially-Invariant Is Not Universal. AAAI Conference on Artificial Intelligence (AAAI), Pages: 6778-6785, February, 2020, New York City, USA. (CCF A)

[16] Huiting Hong, **Xin Li***, Mingzhong Wang: GANE: A Generative Adversarial Network Embedding. IEEE Trans. Neural Networks Learn. Syst. 31(7): 2325-2335 (2020) (SCI—E)

Atari Games

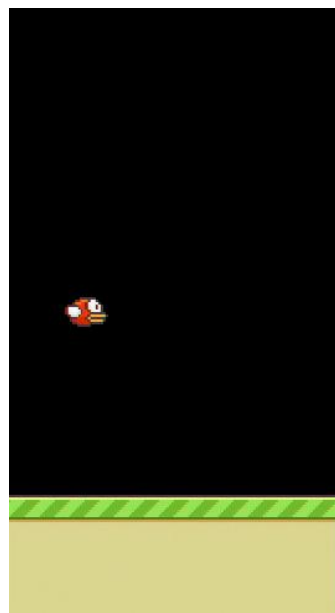


Atari游戏-Pong



Atari游戏-Breakout

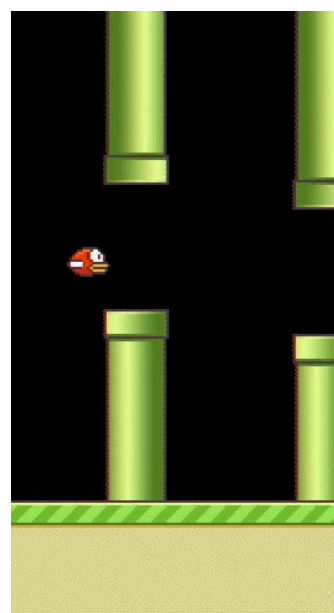
Show Case for Flappy Bird



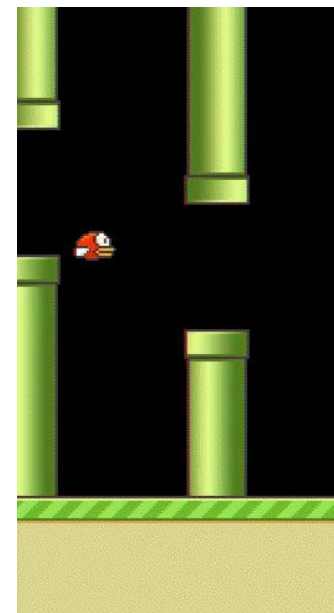
t=25万 2-3



t=100万 24次

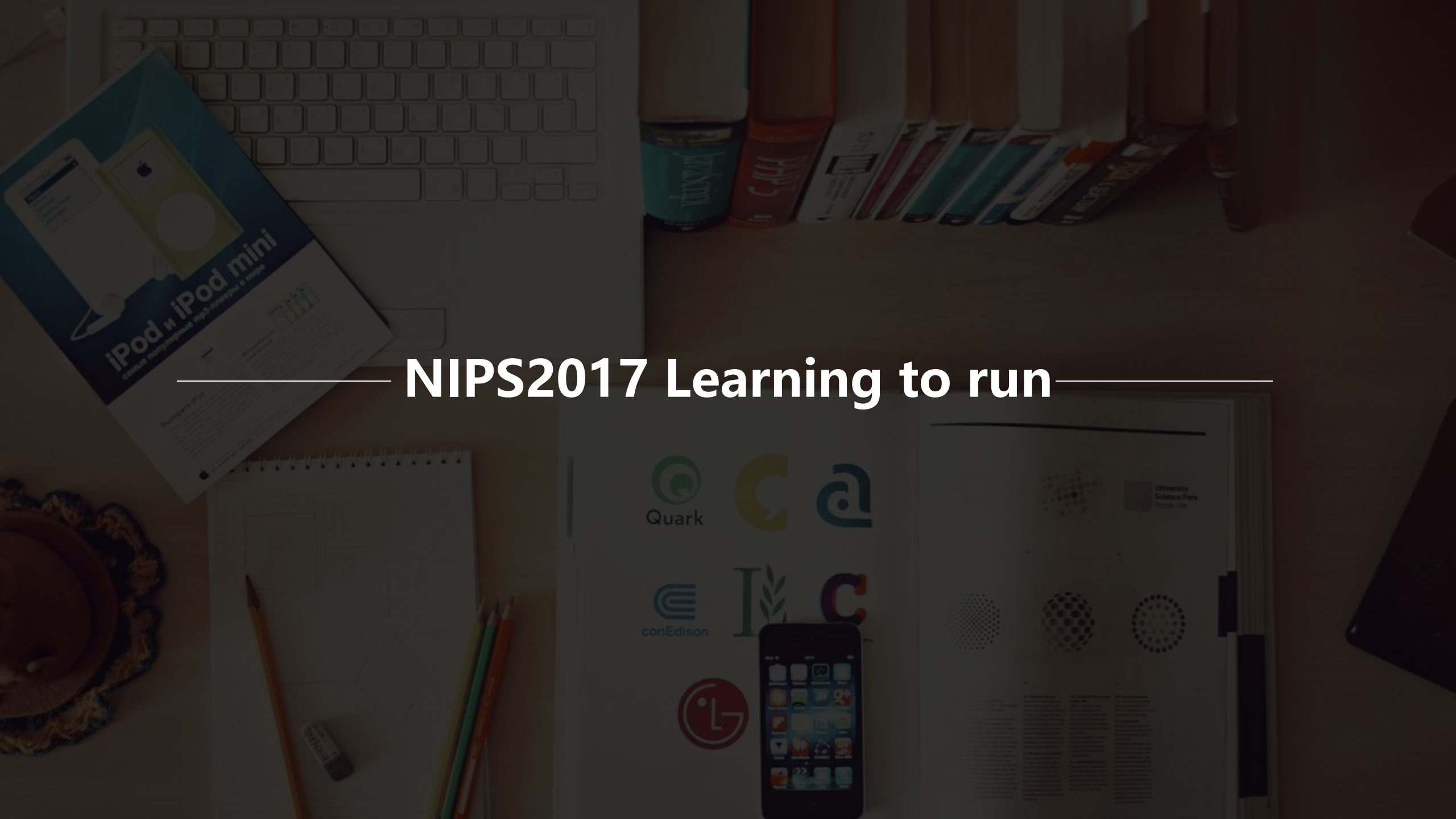


t=210万 死不掉



t=280万 死不掉

次
以上为不同迭代次数下，训练的神经网络能达到的水平。在迭代到200万次以上时，神经网络控制的小鸟已经很难死掉了。

A top-down view of a desk with various items. In the top left, a white laptop keyboard is visible. To its right is a row of colorful books standing upright. Below the keyboard is a magazine titled 'iPod и iPod mini' with a blue and yellow cover. In the bottom left, there is a spiral-bound notebook with a pencil and an eraser. In the bottom center, a smartphone with a grid of app icons is placed on top of a document. The document features several logos: Quark, conEdison, and LG, along with stylized letters 'C', 'a', 'I', and 'C'. To the right of the smartphone is another document with circular patterns and the text 'University Science Park'.

NIPS2017 Learning to run

Environment : OpenSim simulator

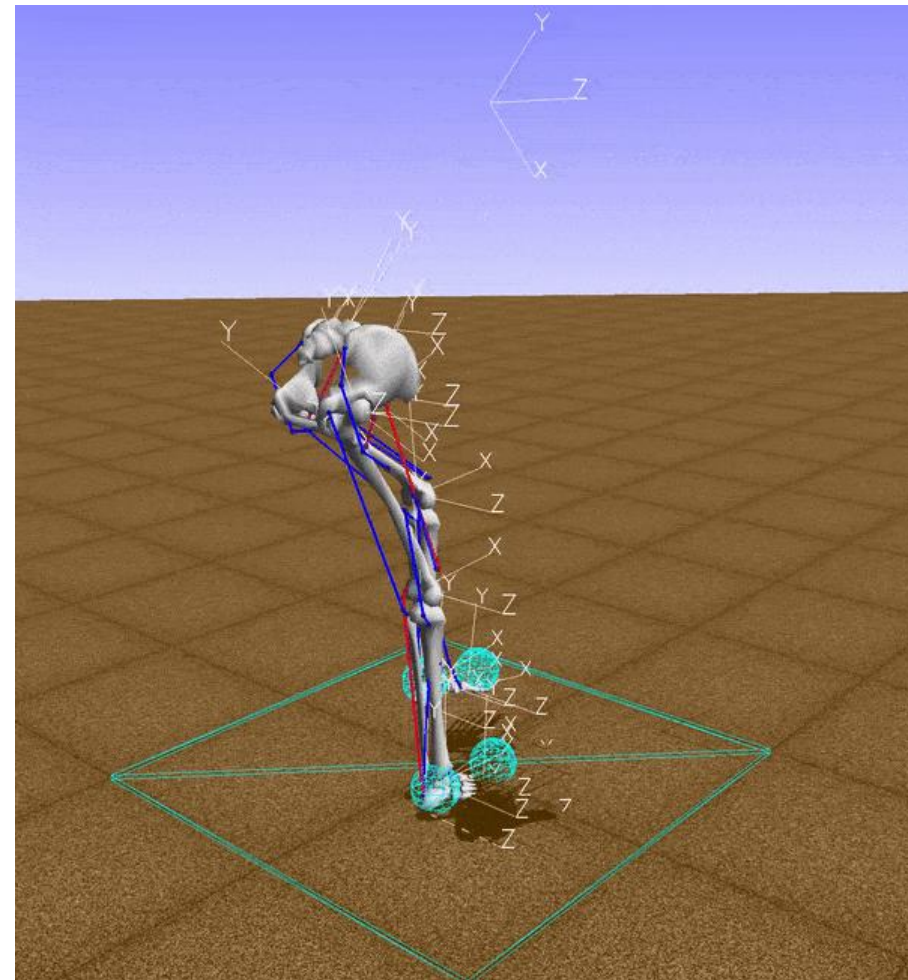
OpenSim人体仿真建模及生物力学分析系统

OpenSim软件是基于C++和JAVA语言为基础开发的一款应用于肌肉模型开发、模拟仿真与分析神经肌肉系统的开放性软件(Simtk.org).

Main objective :

Train physiologically-based human model to navigate a complex obstacle course as quickly as possible. You are provided with a human musculoskeletal model and a physics-based simulation environment where you can synthesize physically and physiologically accurate motion. Potential obstacles include external obstacles like steps, or a slippery floor, along with internal obstacles like muscle weakness or motor noise. You are scored based on the distance you travel through the obstacle course in a set amount of time.

It was also a time limit in the form 1000 maximum timesteps in simulation.



Observation : 41 floats (distance and velocity of various body parts and obstacle locations)

Action : 18 float values from 0 to 1, corresponding to muscle activations

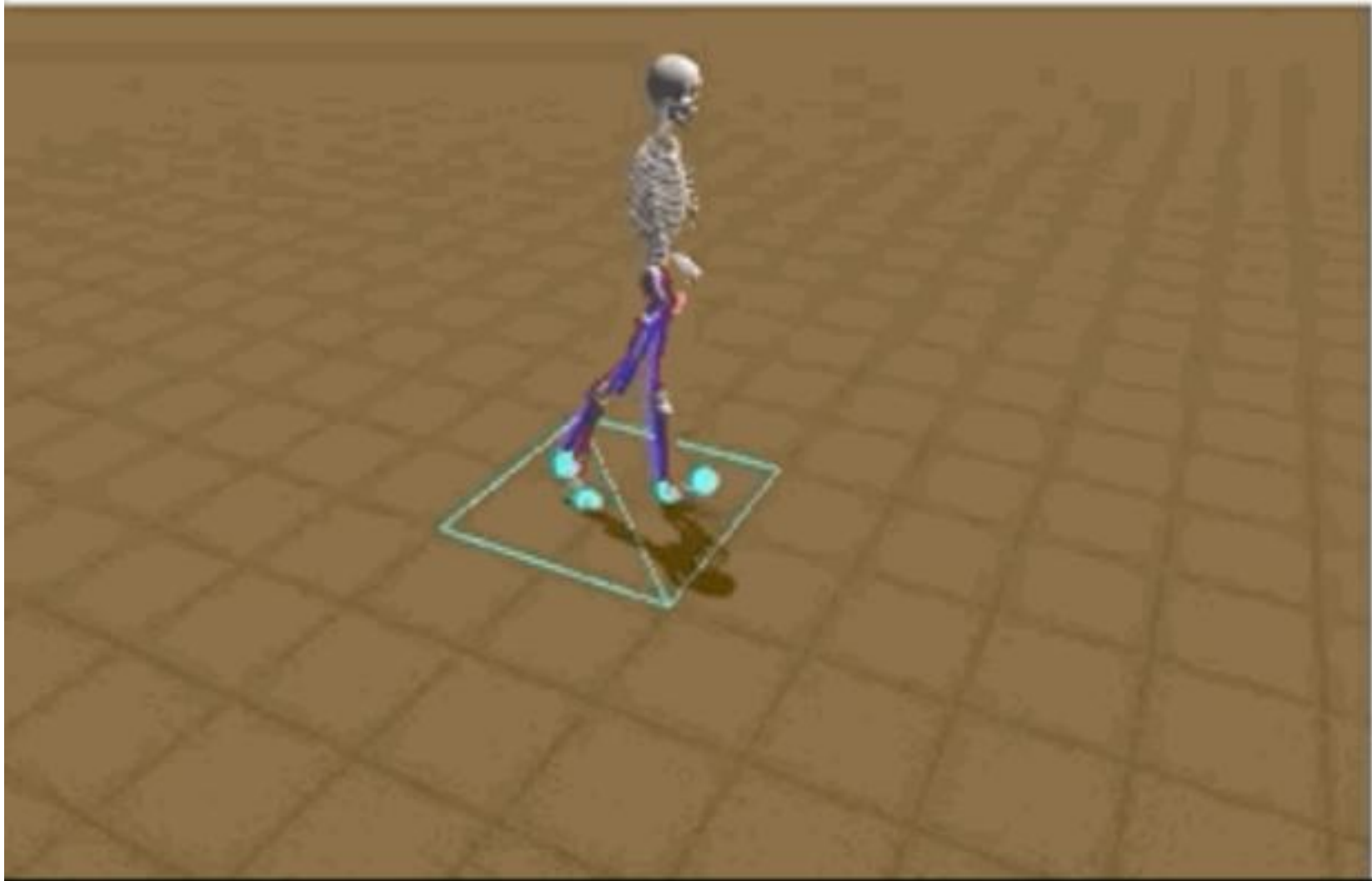
Reward : distance passed plus a small penalty for using ligament forces

➤ Ends if agent falls (pelvis y < 0.65) or after 1000 steps in environment.

Index	English	Index	Name	Index	Name
		0	pelvis.r	22	head.x
		1	pelvis.x	23	head.y
		2	pelvis.y	24~	pelvis.x
		3	pelvis.v.r	25~	pelvis.y
		4	pelvis.v.x	26	torso.x
		5	pelvis.v.y	27	torso.y
		6	hip.right.r	28	toes.left.x
		7	knee.right.r	29	toes.left.y
		8	ankle.right.r	30	toes.right.x
		9	hip.left.r	31	toes.left.y
		10	knee.left.r	32	talus.left.x
		11	ankle.left.r	33	talus.left.y
		12*	hip.right.v.r	34	talus.right.x
		13*	knee.right.v.r	35	talus.right.y
		14*	ankle.right.v.r	36*	psoas.left.strength
		15*	hip.left.v.r	37	psoas.right.strength
		16*	knee.left.v.r	38	next obstacle x distance from pelvis
		17*	ankle.left.v.r	39	next obstacle y position of the center relative to the ground
		18	mass.x	40	next obstacle radius
		19	mass.y		
		20	mass.v.x		
		21	mass.v.y		

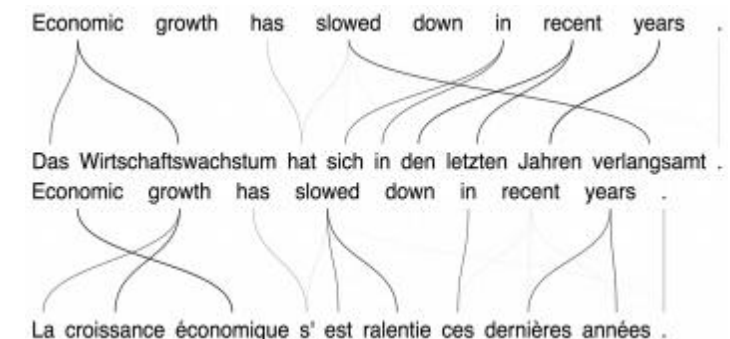
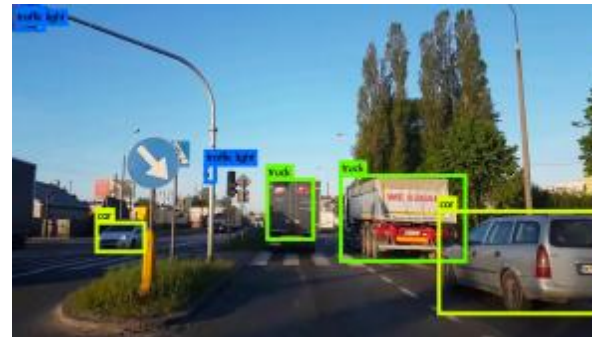
Index	English
0	hamstring
1	biceps femoris
2	gluteus maximus
3	iliopsoas
4	rectus femoris
5	vastus ?
6	gastrocnemius
7	soleus
8	tibialis anterior
9	values of the right leg first

Where:
 r = rotation around z axis
 v = velocity
 * = wrong value because of bug in osim-rl
 ~ = redundant, pelvis is on indices 1-2 already



Machine Learning

- Traditional computer science
 - Program computer for every task
- New paradigm
 - Provide examples to machine
 - Machine learns to accomplish tasks based on examples



Definitions

- Arthur Samuel (1959): **Machine learning** is the field of study that gives computers the ability to learn without being explicitly programmed.
- Tom Mitchell (1998): A computer program is said to **learn** from **experience E** with respect to some class of **tasks T** and performance **measure P**, if its performance at tasks in T, as measured by P, improves with experience E.

Example: Prediction

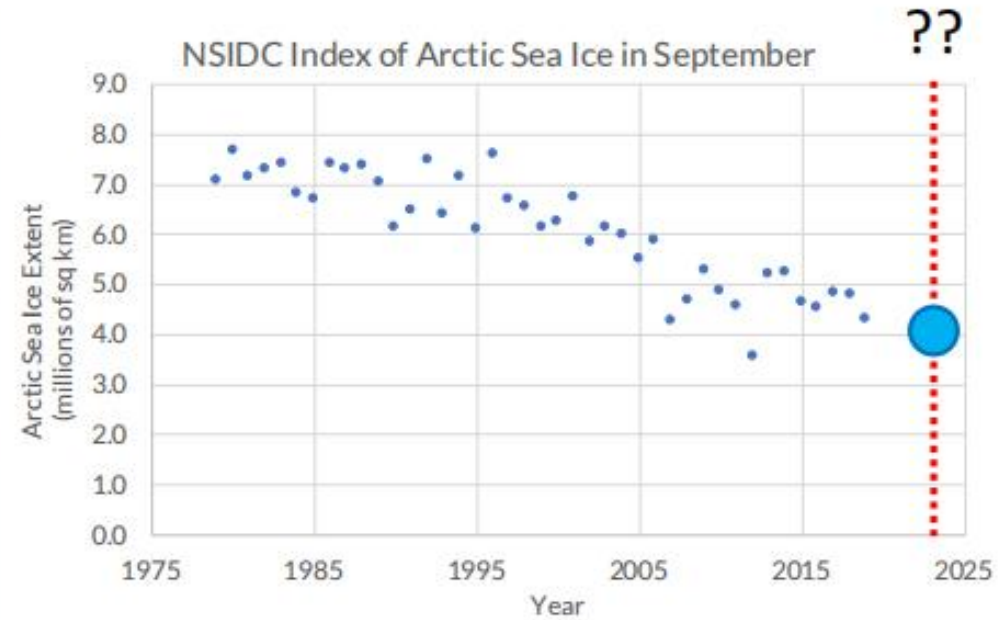


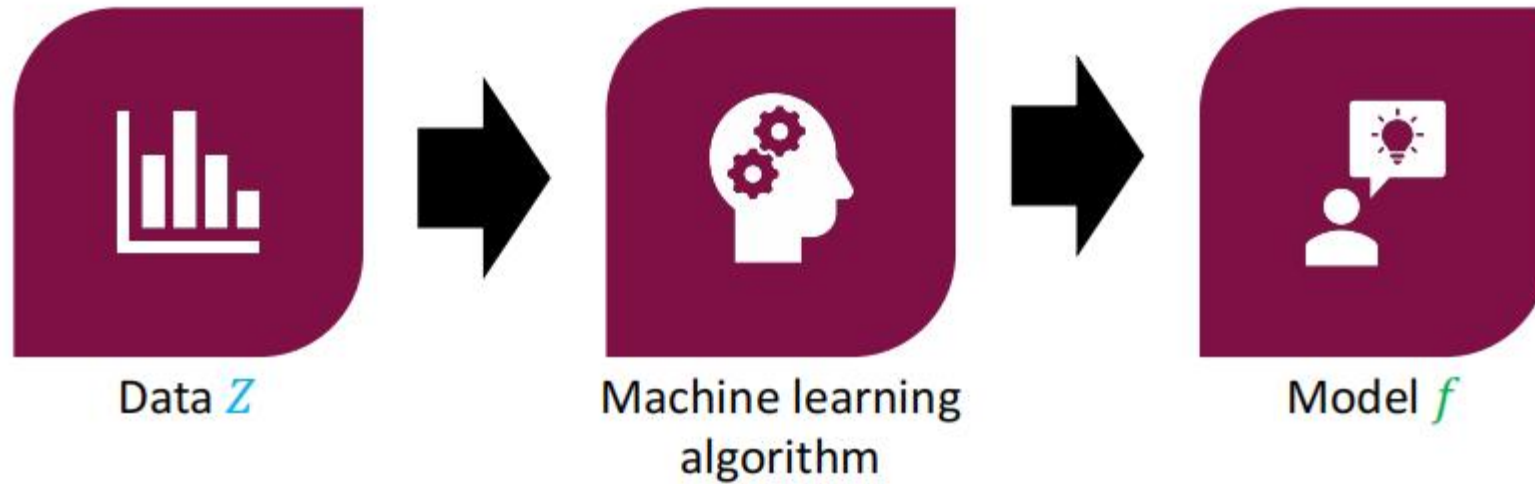
Image: <https://www.flickr.com/photos/gsf/5937599688/>
Data from <https://nsidc.org/arcticseaicenews/sea-ice-tools/>

Example: Prediction

- **Tom Mitchell:** Algorithms that
 - improve their **performance** P
 - at some **task** T
 - with **experience** E
- T = predict Arctic sea ice extent
- P = prediction error (e.g., absolute difference)
- E = historical data



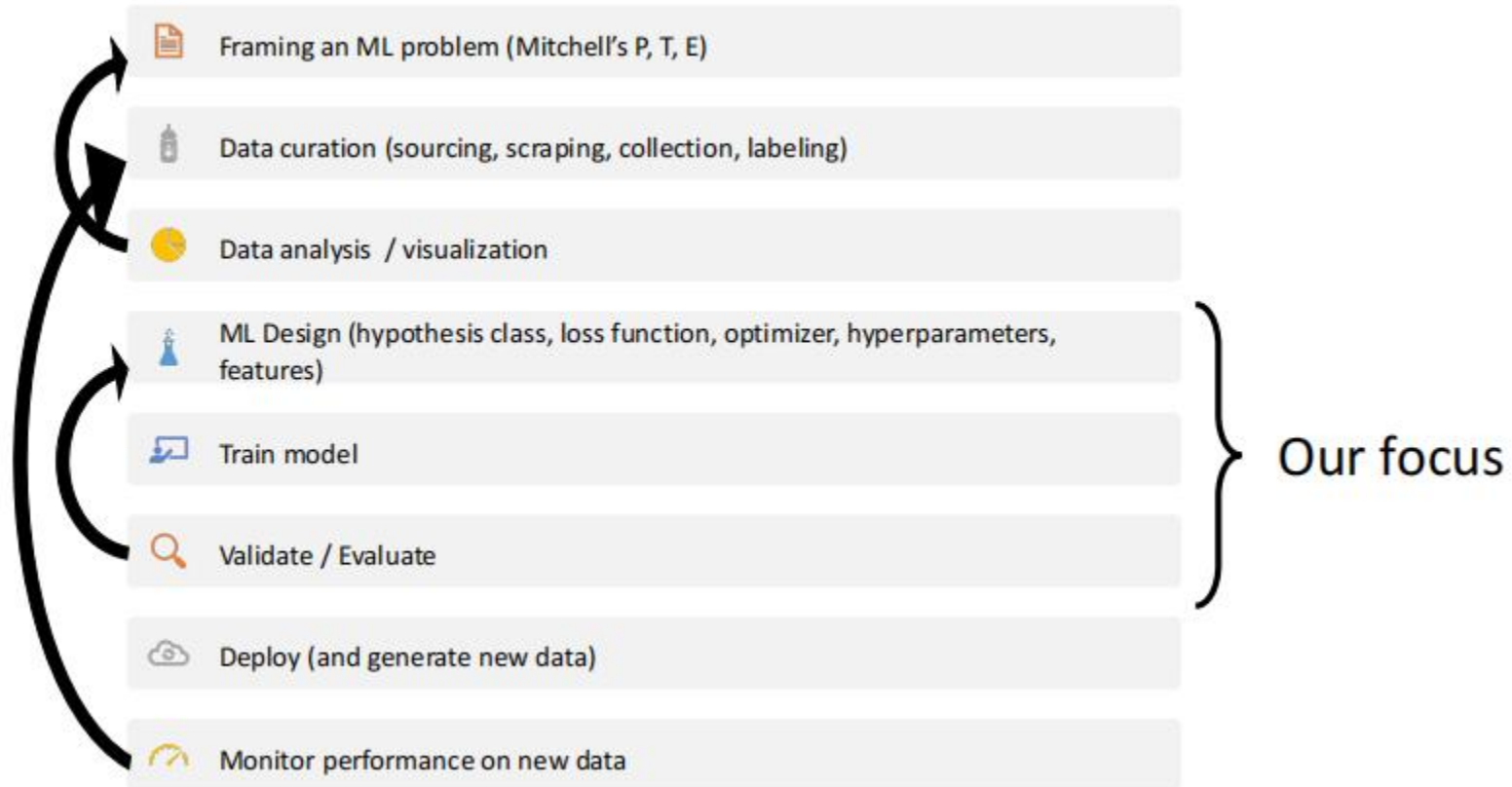
Example: Prediction



Example: Prediction

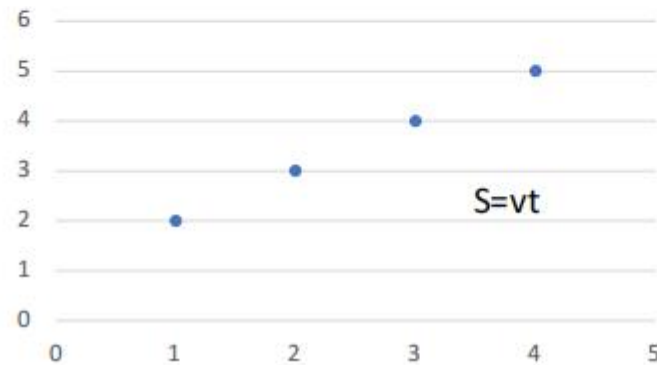


Machine Learning Workflow



Why It Works?

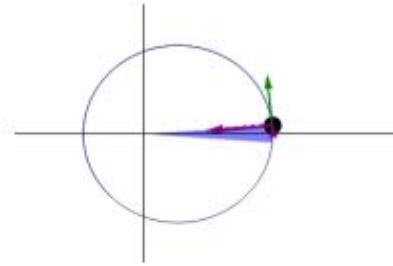
- There are some rules/laws in the data



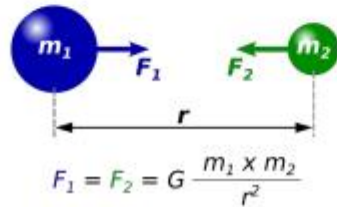
Newton's first law of motion

How Human Learn

- Johannes Kepler found rules from Tycho Brahe's astronomy data (a function to fit the data)
- Kepler's second law



- Newton's law of universal gravitation





Frank Rosenblatt (Cornell)

Linear classifier for character recognition

Based on neurons in the brain

By studying the Perceptron, “the fundamental laws of organization which are common to all information handling systems, machines and men included, may eventually be understood.”

1959

1960

1969



Marvin Minsky, Seymour Papert (MIT)

“Perceptrons: An Introduction to Computational Geometry”

Criticism of Perceptron demonstrating limitations

Research in Perceptrons stops for almost 20 years

Arthur Samuel (IBM)

Checkers playing program

Rote learning- simulate many games to learn strategy

Early reinforcement learning



FIG. 3. Made T Perceptum at Cornell Aeronautical Laboratory. (a) Display unit with sensory input to left, association units in center, and control panel and response units at far right. The sensory to associated subdisplay, shown in (b) is located behind the closed panel to the right of the operator. The image of the letter "C" on the final panel is a separate display, for associating sensory inputs.



1959

1960

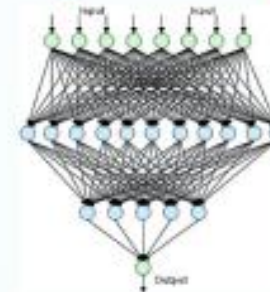
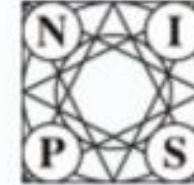
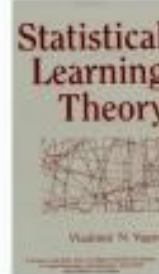
1969

1974

1980

1982

1985



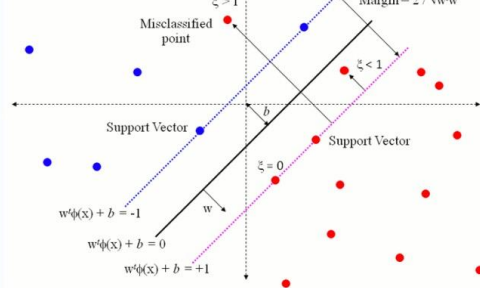
Rise of machine learning as a field

1985- AISTATS

1985- UAI

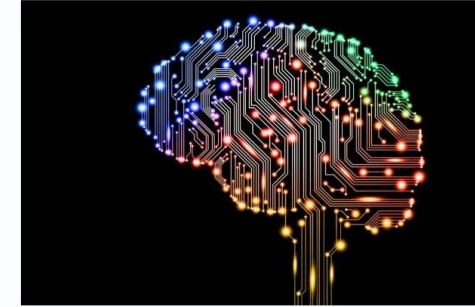
1987-NIPS (90 papers, 2014: 1678 submissions, 414 papers)

1988- COLT



Rise of
the SVMs

3rd rise of
neural
networks
(now deep
learning)



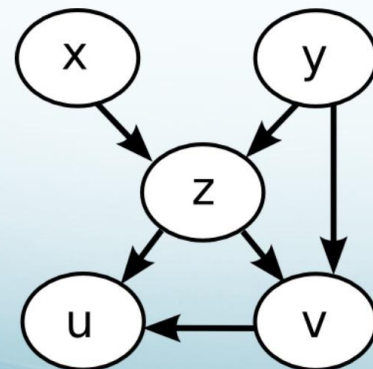
Second
death of
neural
networks

Rise of
the
graphical
models

Deep
learning
takes over

ML hits record size

- 3000 people at NIPS 2015
- ICML 2016 doubles in size



Years are
approximate

2018 Turing Award



For conceptual and engineering breakthroughs that have made deep neural networks a critical component of computing.

ff >

→

←

231 4.5

232 Finally, we compare *DGRL* on a hard vision bas

233 and compare with an existing state of the art ba

234 lation task is adapted from Nasiriany et al. (20

235 superior compared to previous goal conditionin

236 DoF robotic arm from image input and move a

237 designed for training and generalization, where

238 puck in desired positions in a temporally extend

239 complex motor task, since test time generalizati

Connected Users

- X Xin LI
- A anirudhgoyal9119
- R Riashat Islam
- Y Yoshua Bengio

Review Share Subn

Types of Learning

- **Supervised learning**
 - **Input:** Examples of inputs and desired outputs
 - **Output:** Model that predicts output given a new input
- **Unsupervised learning**
 - **Input:** Examples of some data (no “outputs”)
 - **Output:** Representation of structure in the data
- **Reinforcement learning**
 - **Input:** Sequence of interactions with an environment
 - **Output:** Policy that performs a desired task

Three Categories



Supervised learning



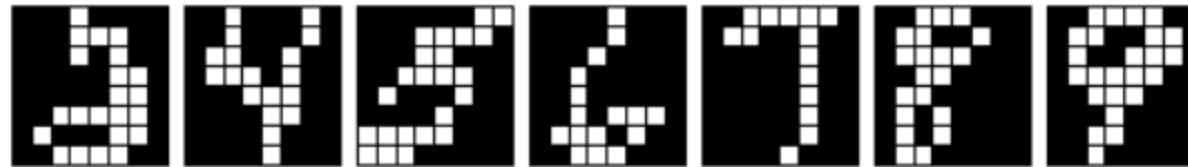
Reinforcement learning



Unsupervised learning

Supervised Learning

- Example: digit recognition (postal code)

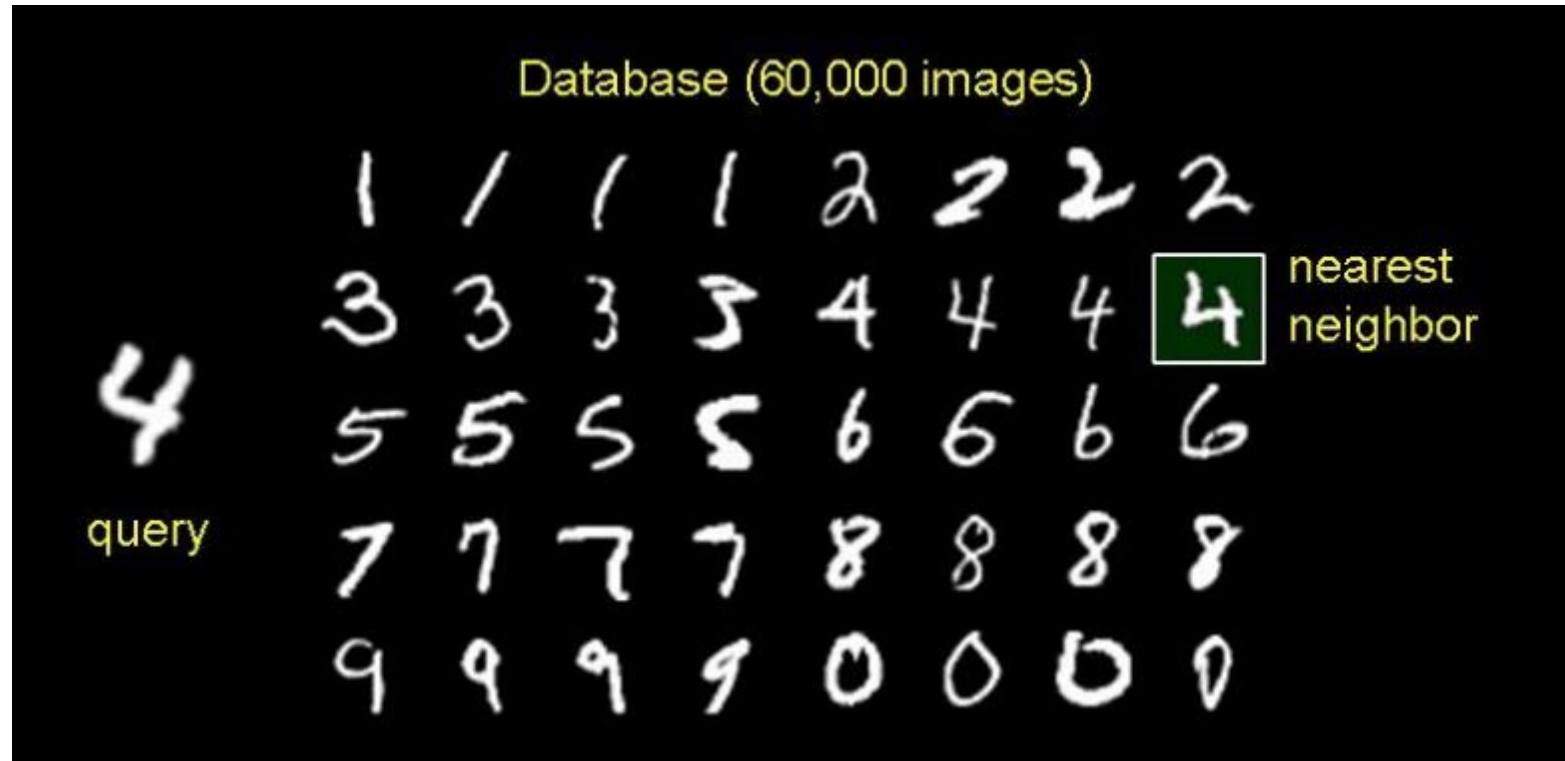


- Simplest approach:
memorization

0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

Supervised Learning

- Nearest neighbour:



Supervised Learning

- Given $(x_1, y_1), \dots, (x_n, y_n)$, learn a function that predicts y given x
- **Classification:** Labels y are categories

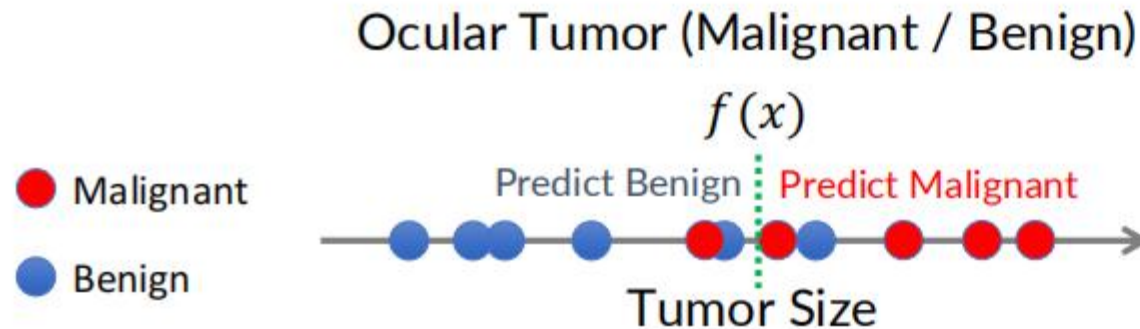
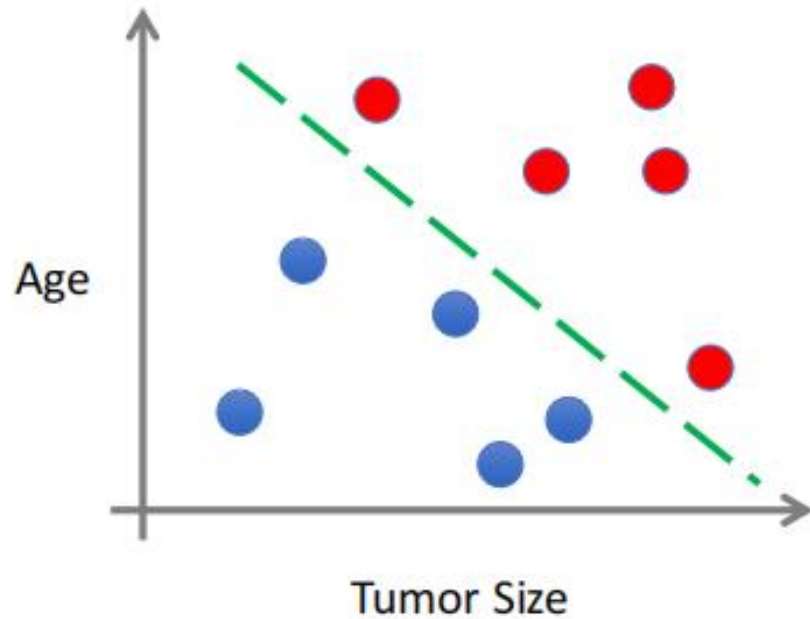


Image: <https://eyecancer.com/uncategorized/choroidal-metastasis-test/>

Supervised Learning

- Given $(x_1, y_1), \dots, (x_n, y_n)$, learn a function that predicts y given x
- Inputs x can be multi-dimensional



- Patient age
- Clump thickness
- Tumor Color
- Cell type
- ...



ImageNet Classification

- 1000 classes
- 1 million images
- Deep neural networks
(supervised learning)



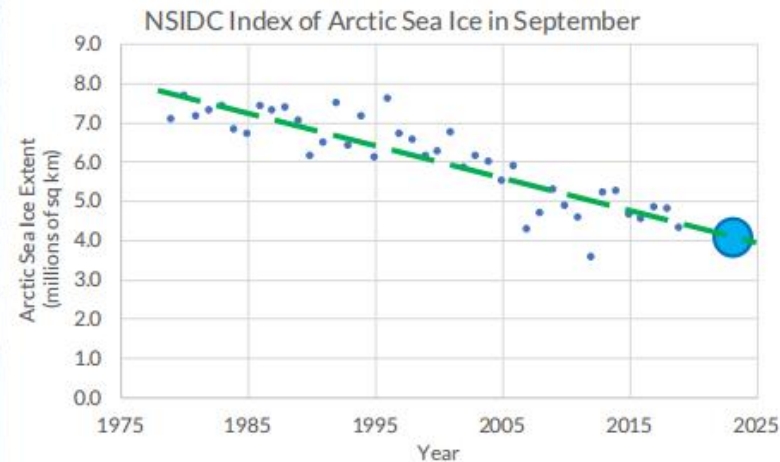
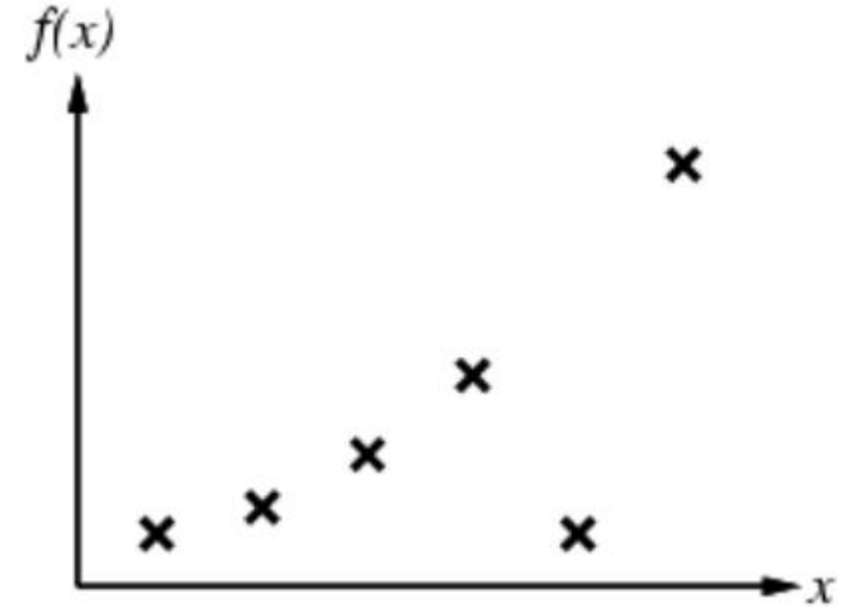
More Formally

- Inductive learning (for supervised learning):
 - Given a **training set** of **examples** of the form $(x, f(x))$
 - x is the input, $f(x)$ is the output
 - Return a function h that approximates f
 - h is called the hypothesis

Prediction

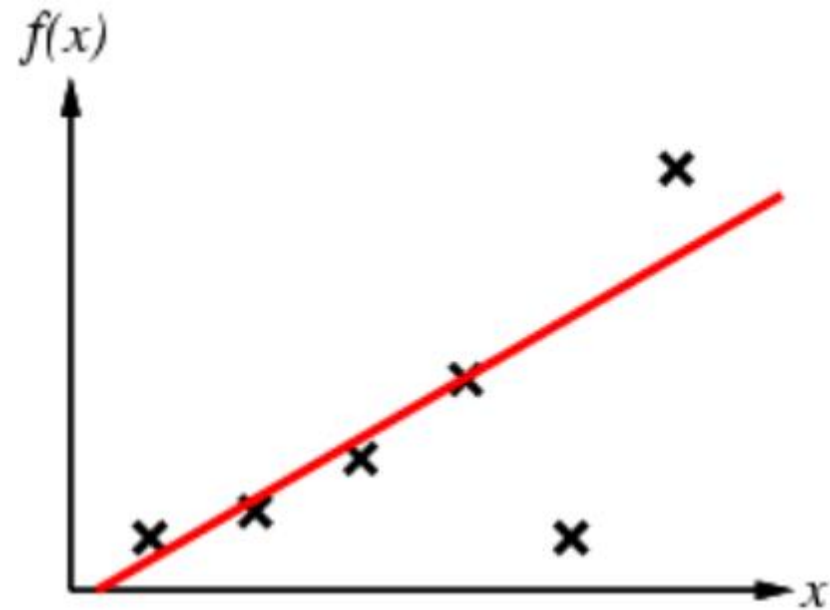
- Find function h that fits f at instances x

- Given $(x_1, y_1), \dots, (x_n, y_n)$, learn a function that predicts y given x
- Regression:** Labels y are real-valued



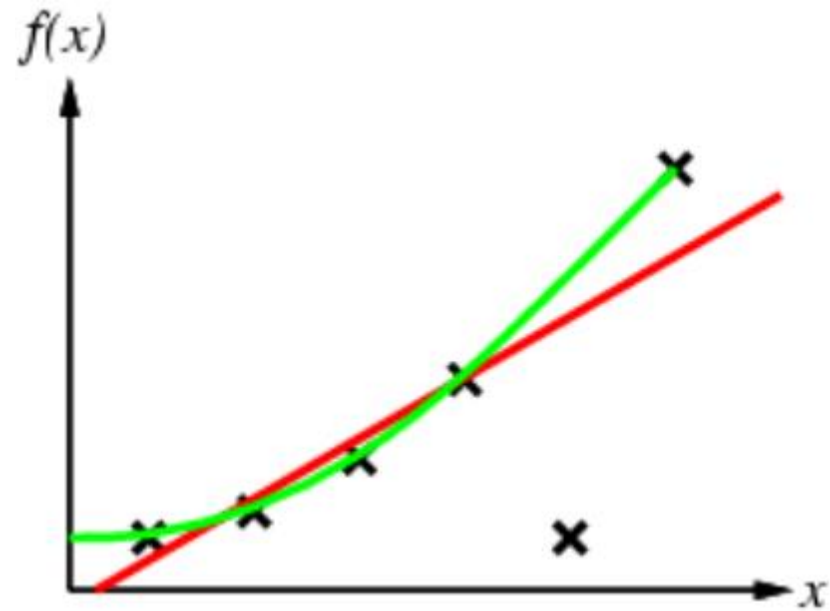
Prediction

- Find function h that fits f at instances x



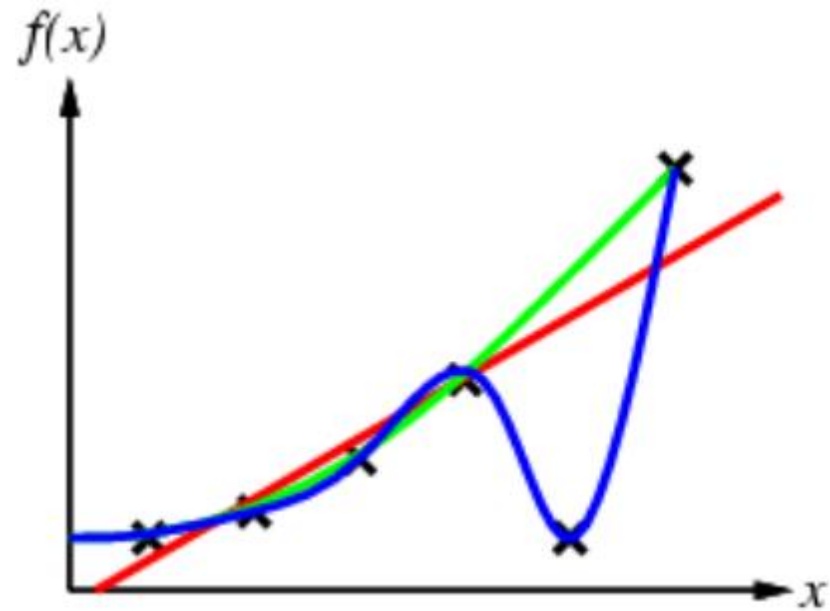
Prediction

- Find function h that fits f at instances x



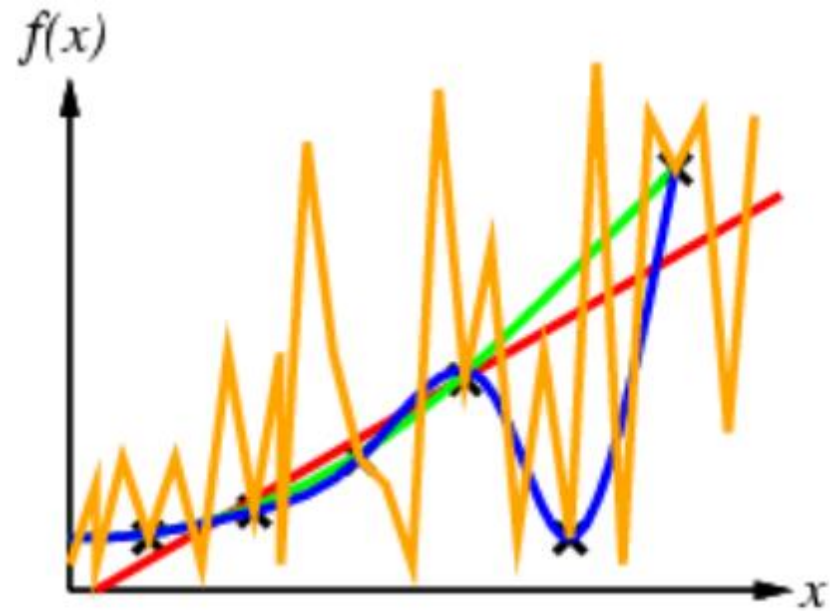
Prediction

- Find function h that fits f at instances x



Prediction

- Find function h that fits f at instances x



Generalization

- Key: a good hypothesis will **generalize well** (i.e., predict unseen examples correctly)
- **Ockham's razor**: prefer the simplest hypothesis consistent with data

奥卡姆剃刀

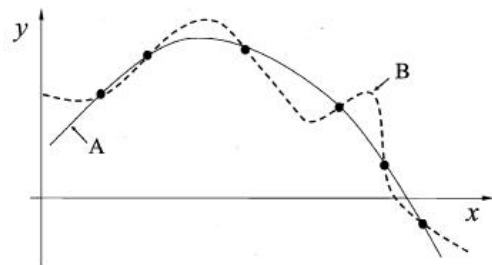
选取偏好的一般性原则：“奥卡姆剃刀”

若有多个假设与观察一致，则选最简单的那个

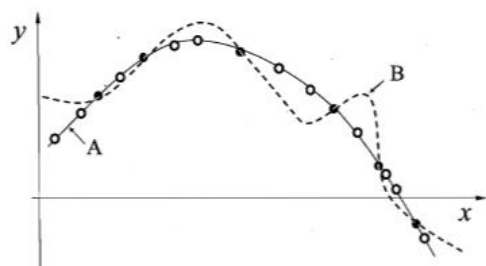
奥卡姆剃刀原则主张选择与经验观察一致的最简单假设，它在自然科学如物理学、天文学等领域中广为沿用。

哥白尼坚持的“日心说”的理由之一就是它比托勒密的“地心说”更加简单且符合观测。

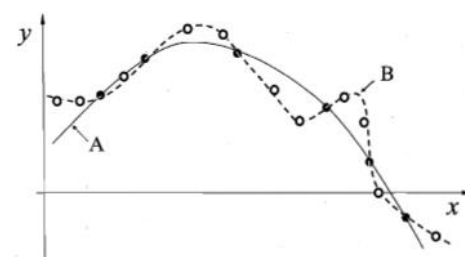
没有免费的午餐



存在多条曲线与样本集一致



A优于B



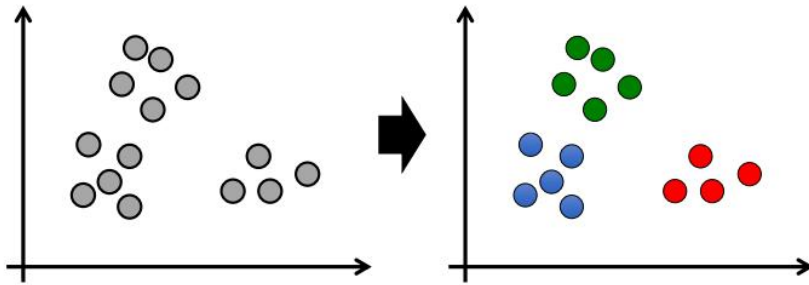
B优于A

在所有可能均等的情况下，任何学习算法在非训练集上的准确率的期望相同。

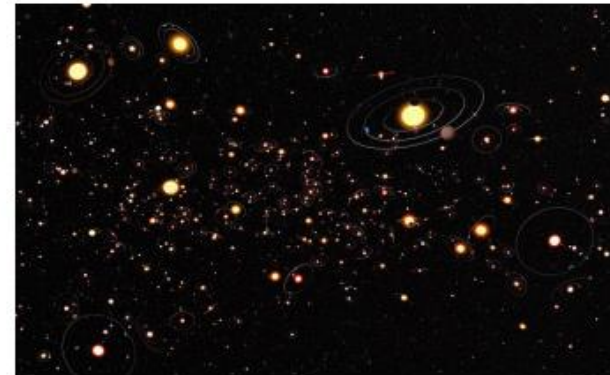
即没有一个学习算法可以在任何领域总是产生最准确的学习器。

Unsupervised Learning

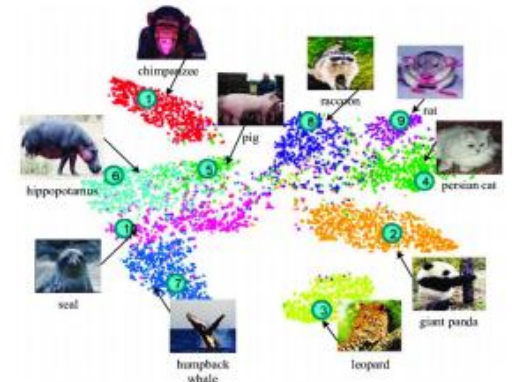
- Given $x_1, \dots, x_n$ (no labels), output hidden structure in x 's
 - E.g., clustering



Find Subgroups in
Social Networks



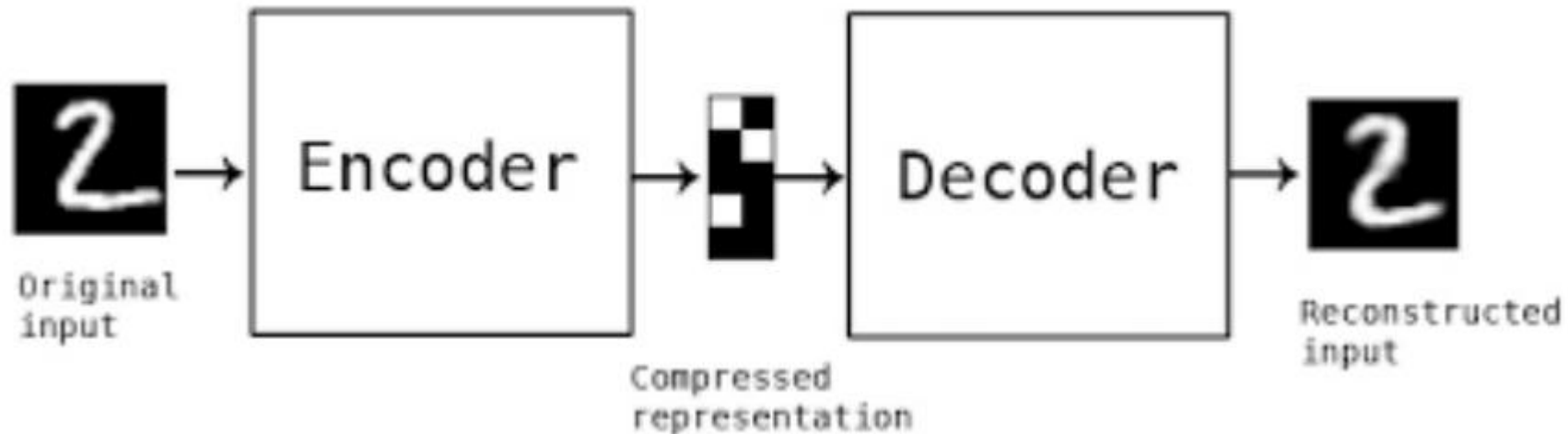
Identify Types of Exoplanets



Visualize Data

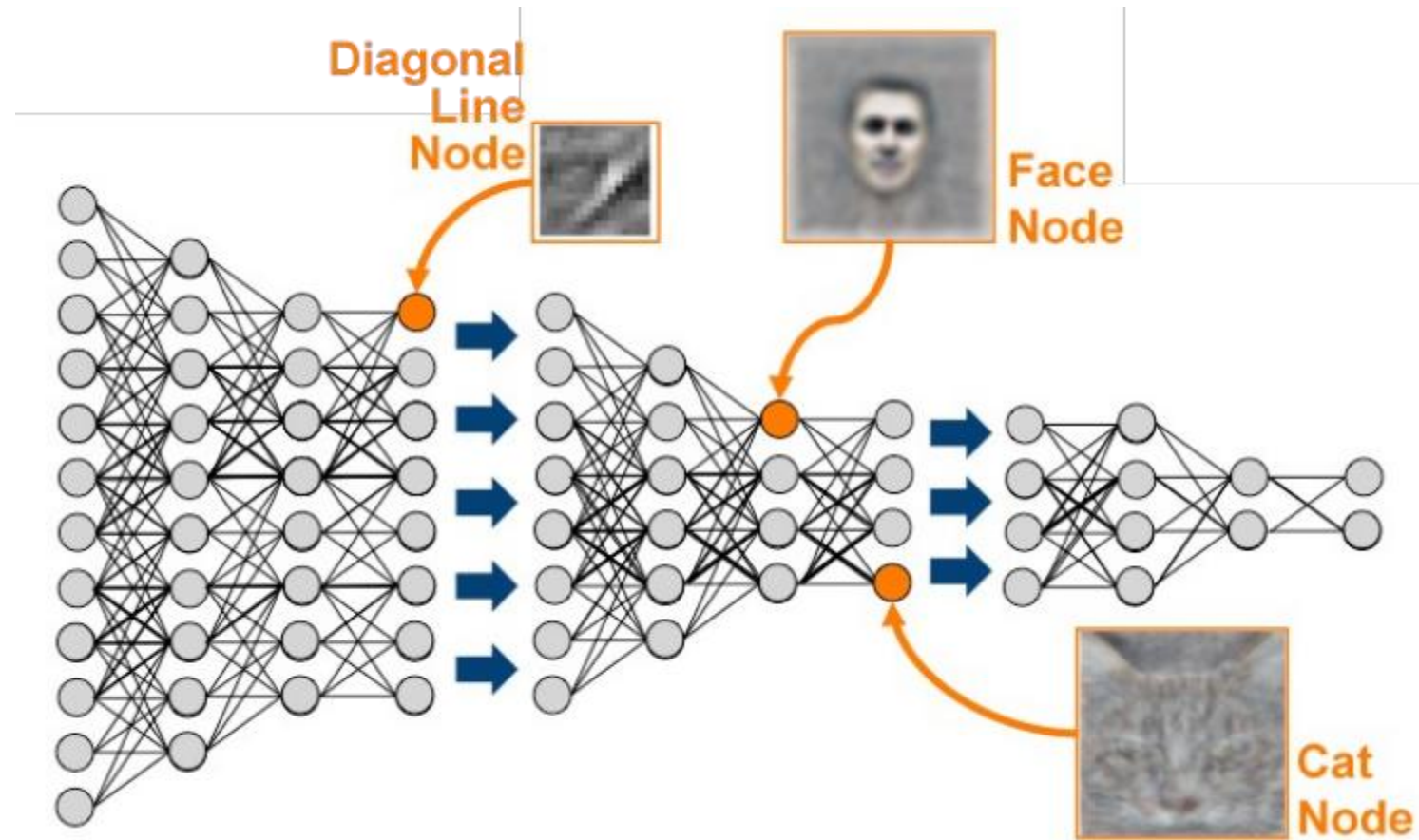
Unsupervised Learning

- Output is not given as part of training set
- Find model that explains the data
 - E.g. clustering, compressed representation, features, generative models



Unsupervised Feature Generation

- Encoder trained on large number of images



Unsupervised Image Generation

- Which images are real? And which ones are fake?

Real



CelebA (Liu et al., 2015)

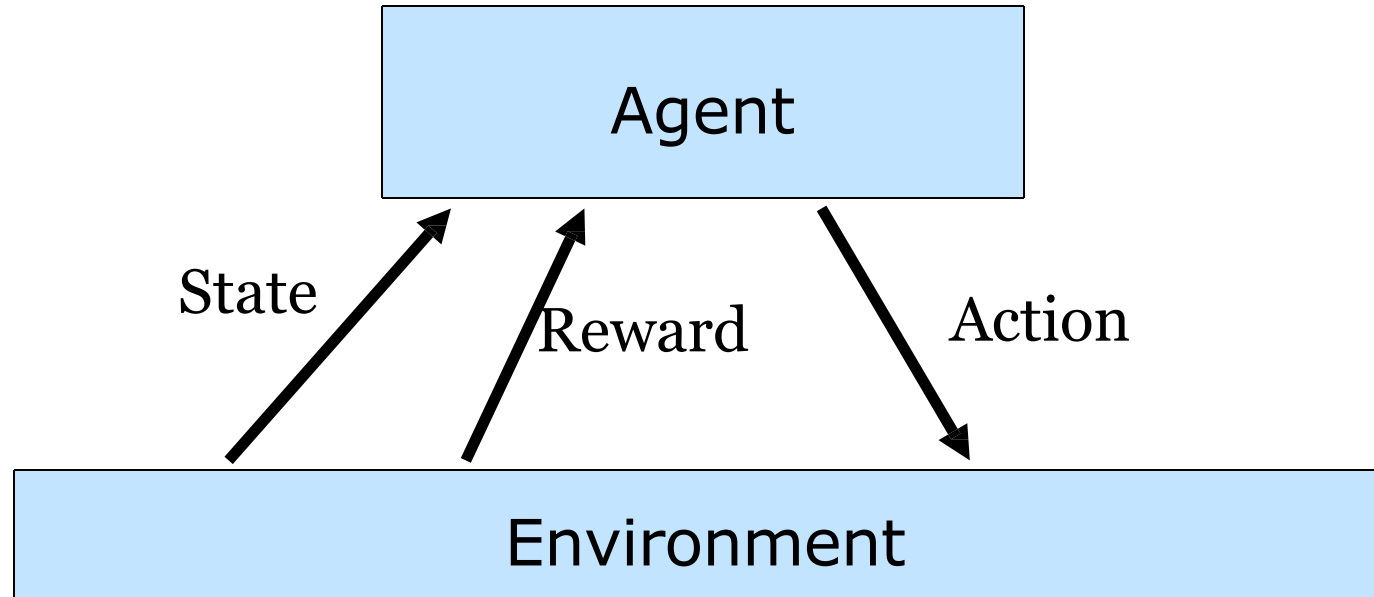


Fake!

StyleGAN2 (Karras et al., 2020)

- Image generation: variational autoencoders, generative adversarial networks, diffusion models

Reinforcement Learning Problem



Goal: Learn to choose actions that maximize rewards

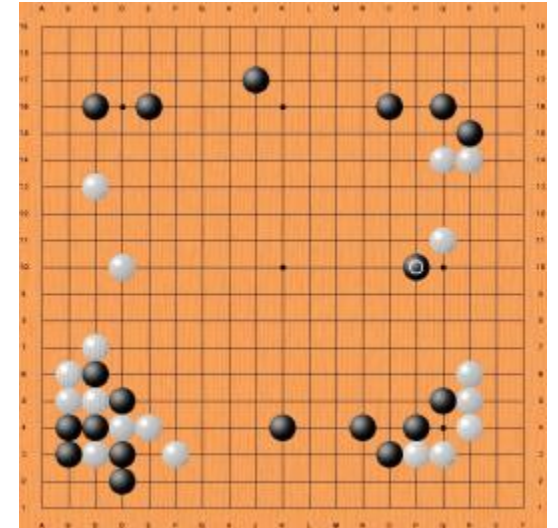
Animal Psychology

- Negative reinforcements:
 - Pain and hunger
 - Positive reinforcements:
 - Pleasure and food
 - Reinforcements used to train animals
-
- Let's do the same with computers!



Game Playing

- Example: Go (one of the oldest and hardest board games)
- **Agent:** player
- **Environment:** opponent
- **State:** board configuration
- **Action:** next stone location
- **Reward:** +1 win / -1 loose



2016: AlphaGo defeats top player Lee Sedol (4-1)
Game 2 move 37: AlphaGo plays unexpected move (odds 1/10,000)

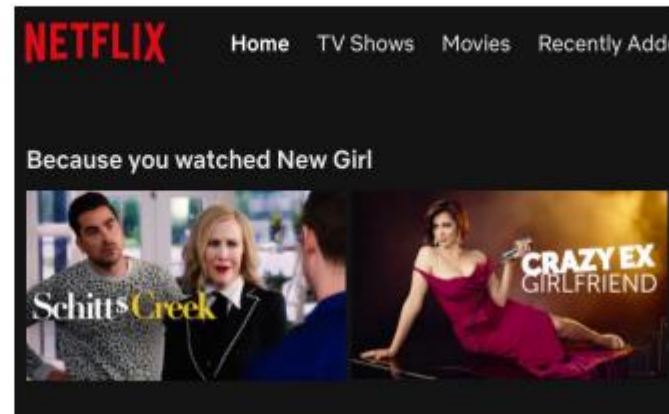
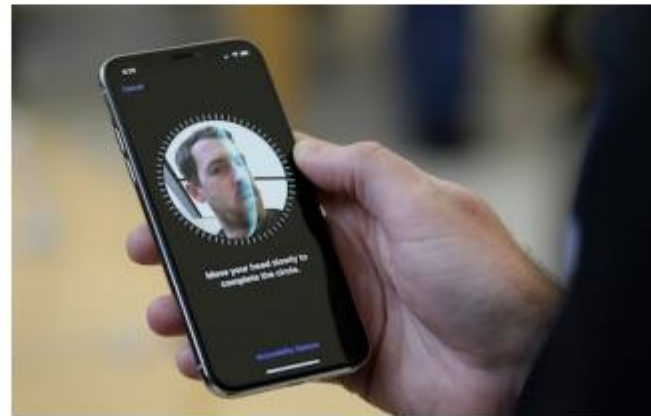
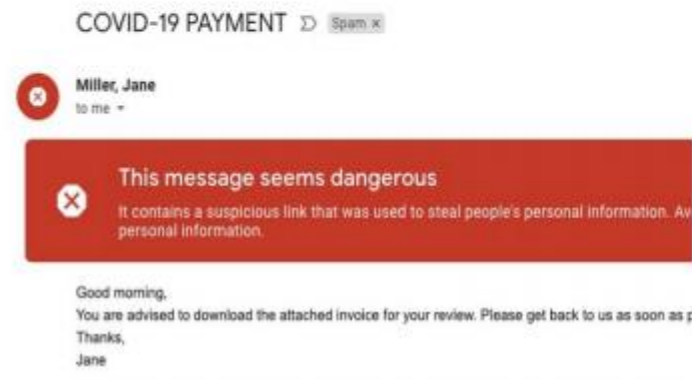
Combining Unsupervised, Supervised and Reinforcement Learning

- Modern systems:
 - Phase 1: unsupervised feature extraction (no labels)
 - Phase 2: supervised training (exploit labels)
 - Phase 3: fine tune by reinforcement learning (exploit reinforcements)
- Alpha Go: supervised + reinforcement learning
- Sentiment analysis with BERT: unsupervised + supervised learning
- ChatGPT: supervised + reinforcement learning

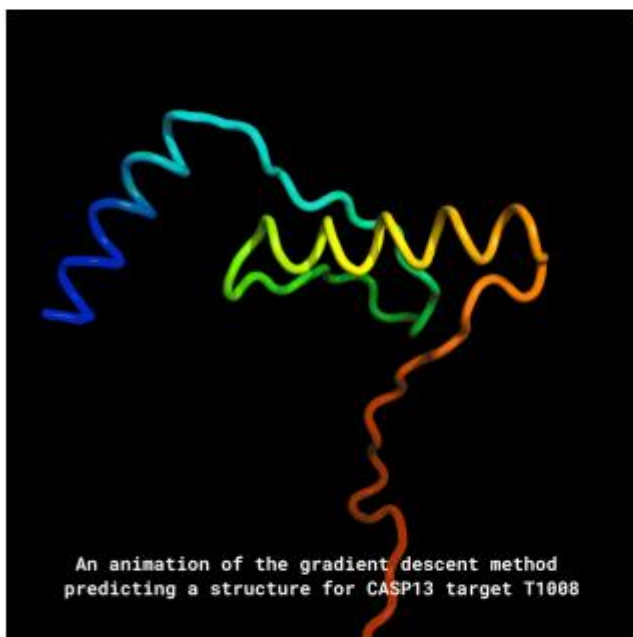
Applications of Machine Learning

- Speech recognition: Siri, Cortana
- Natural Language Processing: Machine translation, dialog systems
- Computer vision: Image and video analysis
- Robotic Control: Autonomous vehicles
- Intelligent assistants: Activity recognition, recommender systems
- Computational finance: Stock trading, portfolio optimization

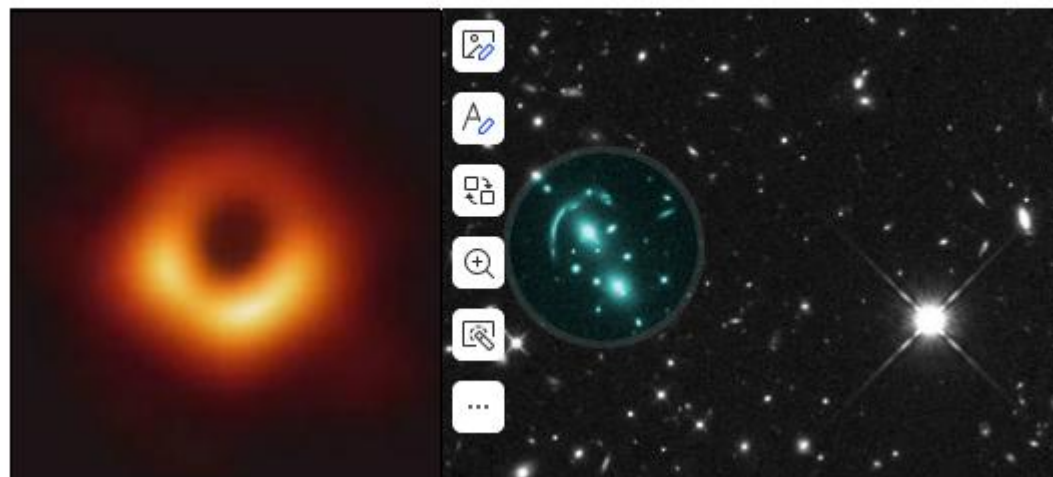
Everyday Applications



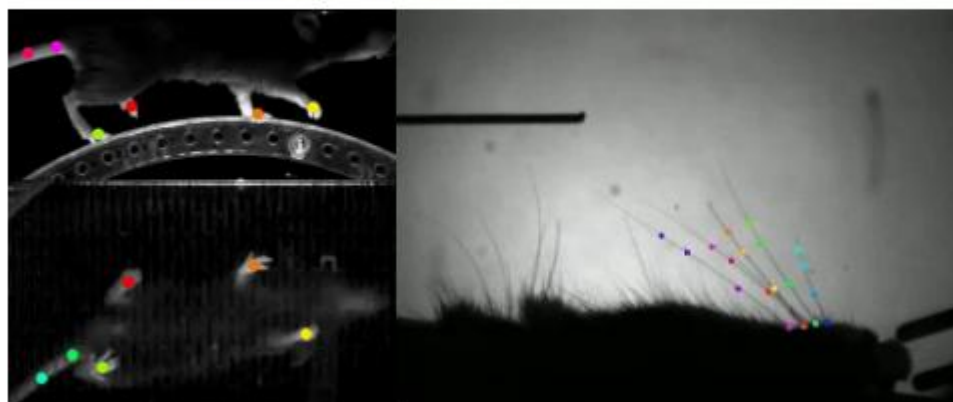
Scientific Discovery



<https://deepmind.com/blog/article/AlphaFold-Using-AI-for-scientific-discovery>



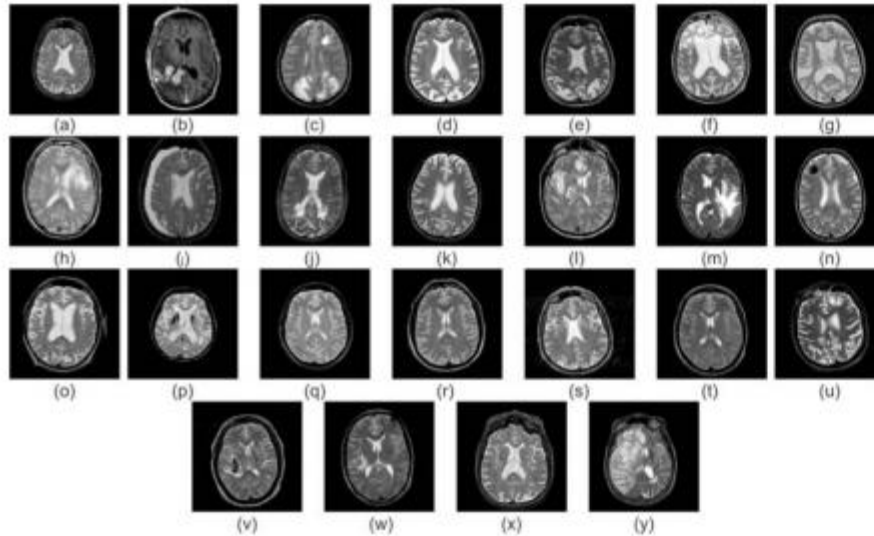
<https://www.jpl.nasa.gov/edu/news/2019/4/19/how-scientists-captured-the-first-image-of-a-black-hole/>



<http://www.mousemotorlab.org/deeplabcut>

Radiology and Medicine

Input: Brain scans



Output: Neurological disease labels

Machine learning studies on major brain diseases: 5-year trends of 2014–2018

Koji Sakai¹ · Kei Yamada¹

Applications of machine learning in drug discovery and development

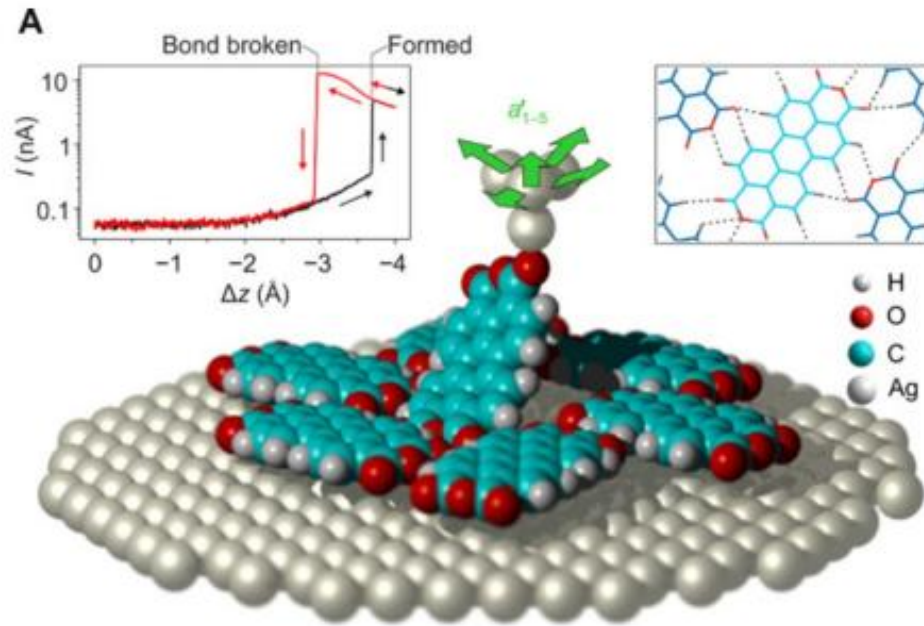
<https://www.nature.com/articles/s41573-019-0024-5>

Deep learning-enabled medical computer vision

Andre Esteva , Katherine Chou, Serena Yeung, Nikhil Naik, Ali Madani, Ali Mottaghi, Yun Liu, Eric Topol, Jeff Dean & Richard Socher

<https://www.nature.com/articles/s41746-020-00376-2>

Nanofabrication



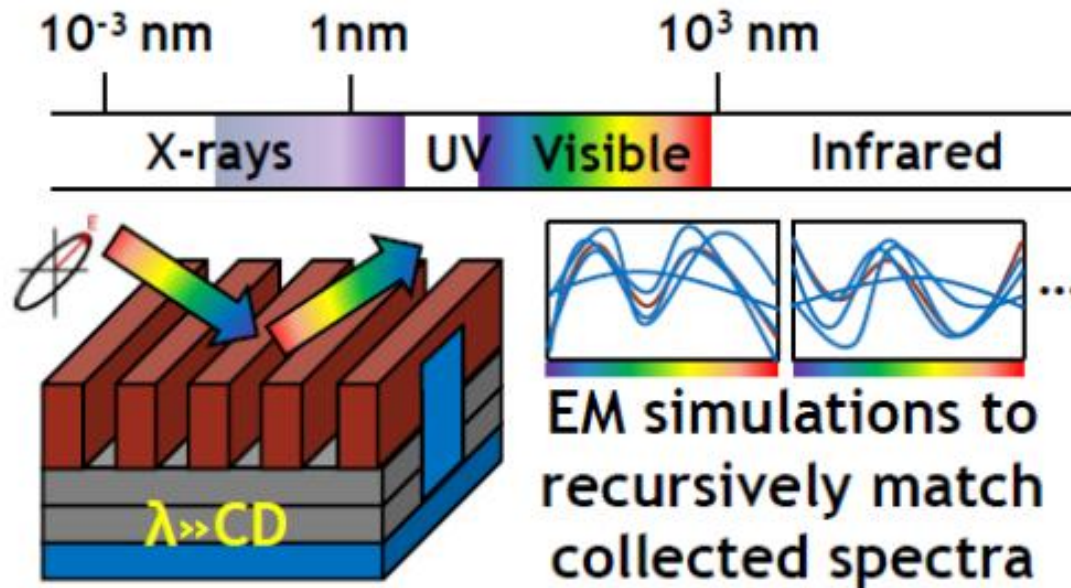
Autonomous robotic nanofabrication with reinforcement learning

Removing individual molecules autonomously with a scanning probe microscope to manufacture new material

RL agent does better than humans!

Semiconductor Manufacturing

Input: light spectra after bouncing off silicon wafer



Output: defective / perfect

huge gains in manufacturing
throughput
→ cheaper phones and computers!

Text Generation



我是 DeepSeek, 很高兴见到你!

我可以帮你写代码、读文件、写作各种创意内容, 请把你的任务交给我吧~

给 DeepSeek 发送消息

 深度思考 (R1)

 联网搜索



Ethical Considerations

“The Pennsylvania Board of Probation and Parole has begun using machine learning forecasts to help inform parole release decisions. In this paper, we evaluate the impact of the forecasts on those decisions and subsequent recidivism.”

An impact assessment of machine learning risk forecasts on parole board decisions and recidivism

[Richard Berk](#) 

“In 2013, the University of Texas at Austin’s computer science department began using a machine-learning system called GRADE to help make decisions about who gets into its Ph.D. program”

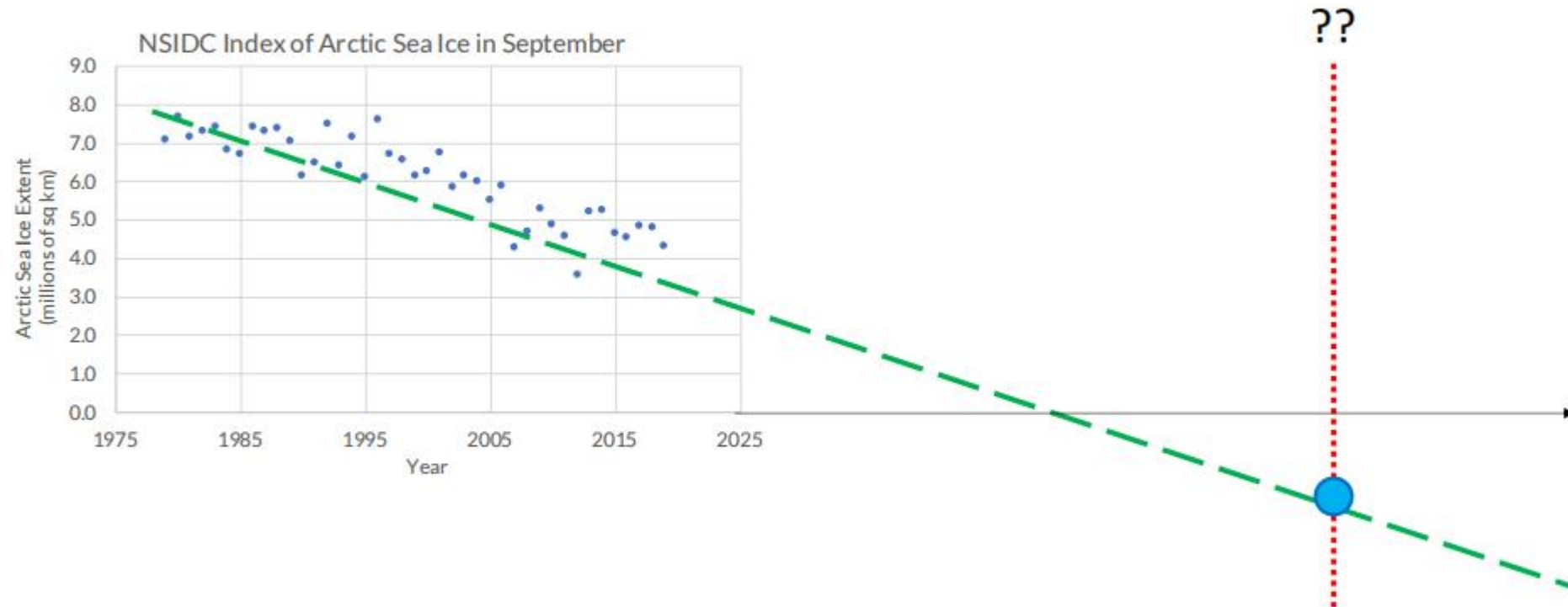
The Death and Life of an Admissions Algorithm

“Videos about vegetarianism led to videos about veganism. Videos about jogging led to videos about running ultramarathons. It seems as if you are never ‘hard core’ enough for YouTube’s recommendation algorithm. It promotes, recommends and disseminates videos in a manner that appears to constantly up the stakes. Given its billion or so users, YouTube may be one of the most powerful radicalizing instruments of the 21st century.”

YouTube, the great radicalizer

THE NEW YORK TIMES / ZEYNEP TUFEKCI / MAR 12

Danger of Out-of-Domain Machine Learning



Any time you are evaluating on data “far” from your training data, beware!

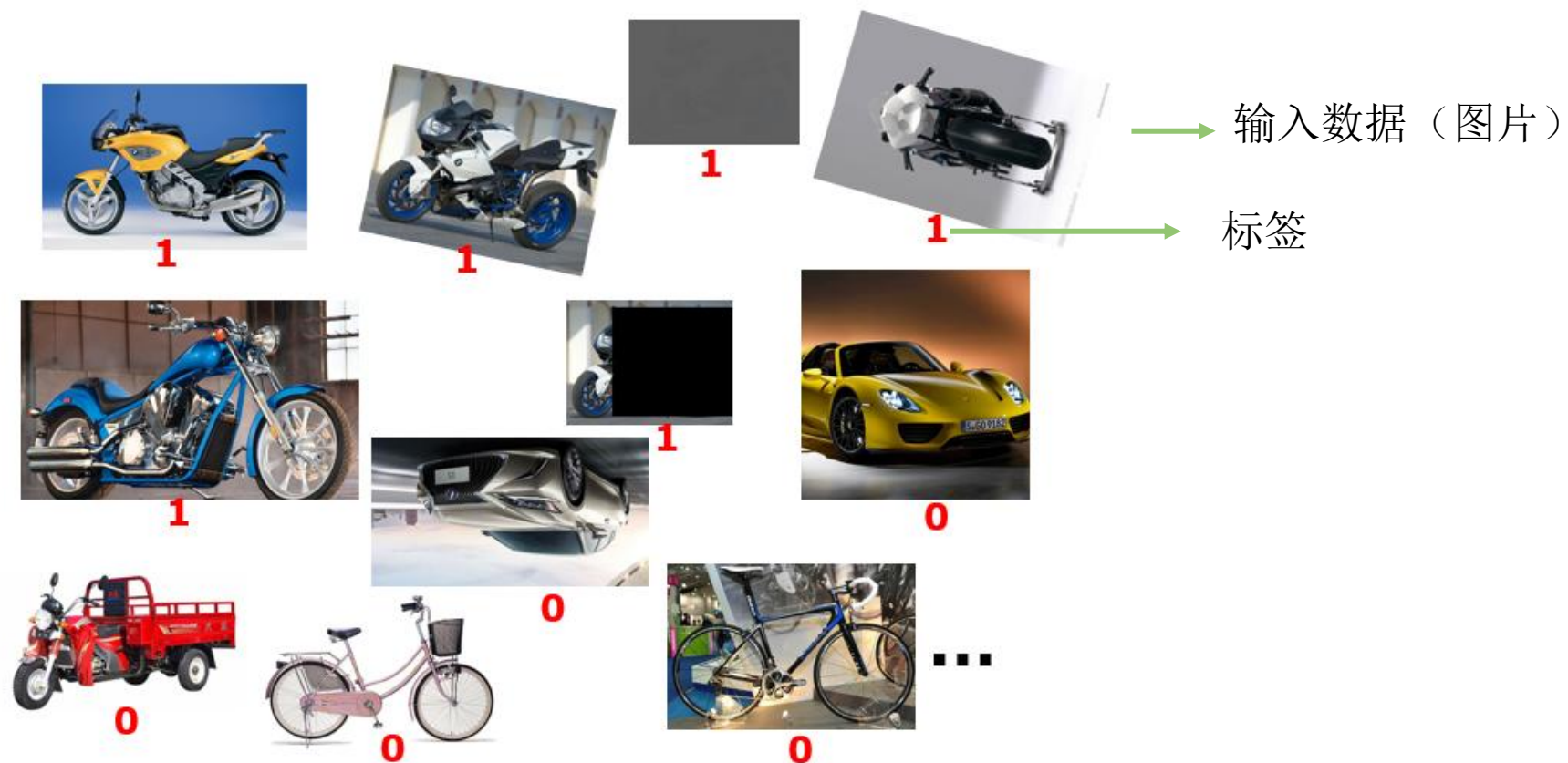
实例

如何判断一幅图上的是否为摩托车？



数据处理

原始数据

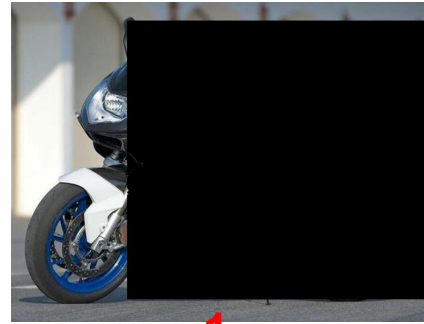


数据处理

□ 清除噪声



1



1

□ 类别平衡



1



1



1



1



0



0



0



0

数据处理

□ 规整数据格式（统一大小并灰度化）



□ 数据增强（可能）



□

数据与特征

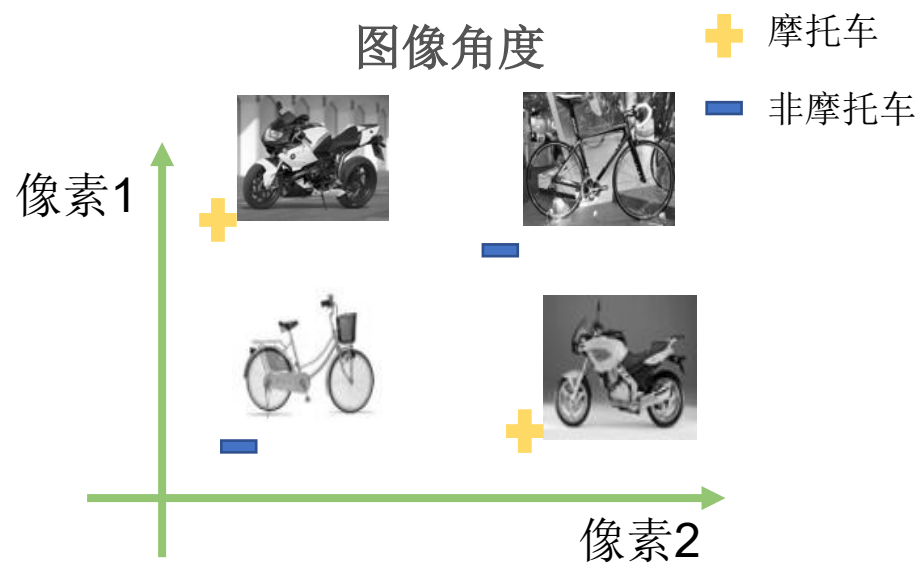
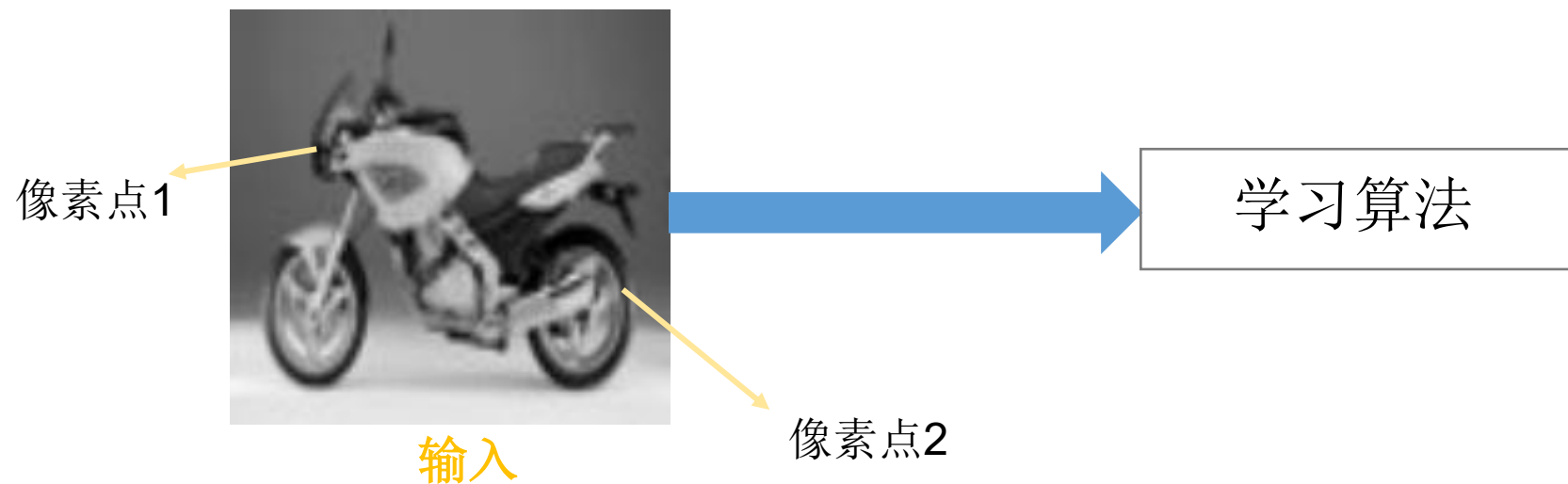
我们眼中的图像：



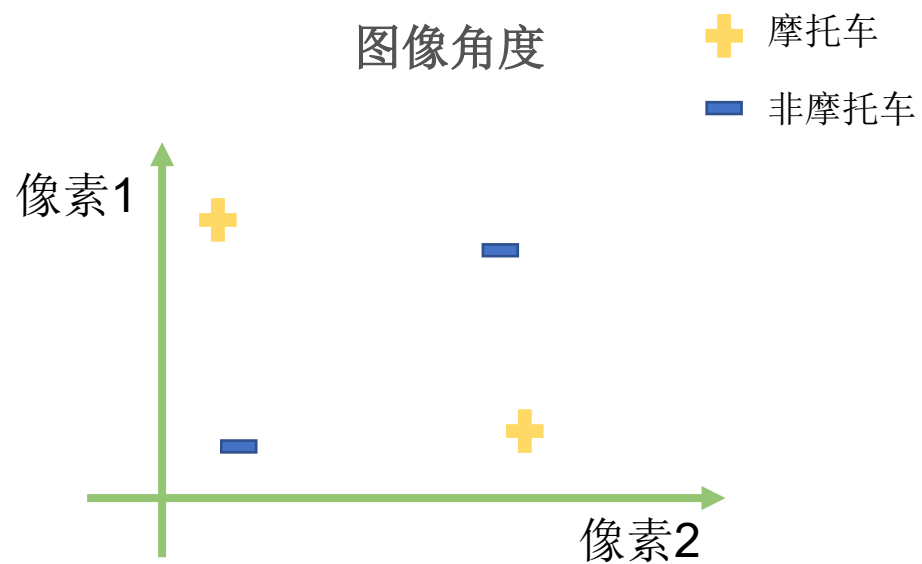
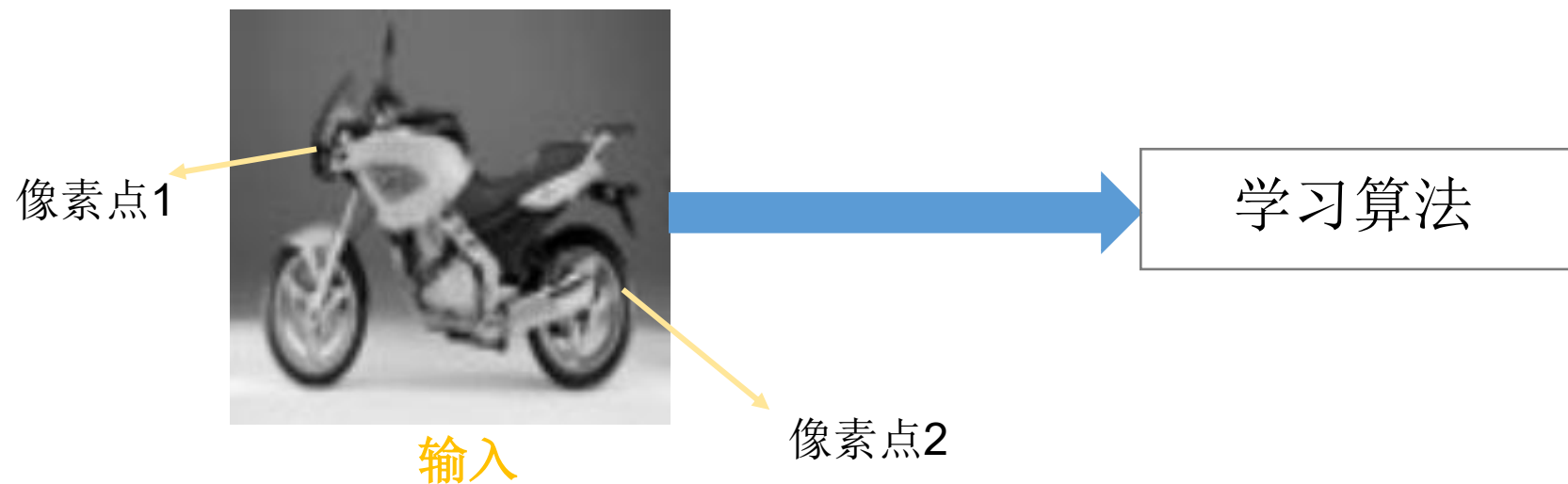
计算机眼中的图像：

212	199	213	215	195	178	158
221	209	205	191	167	147	115
188	176	165	152	140	170	106
154	143	142	149	153	173	101
131	122	138	152	147	128	84
104	105	124	129	113	107	87
98	89	92	98	95	89	88
99	63	45	60	82	58	76
75	56	41	51	73	55	70

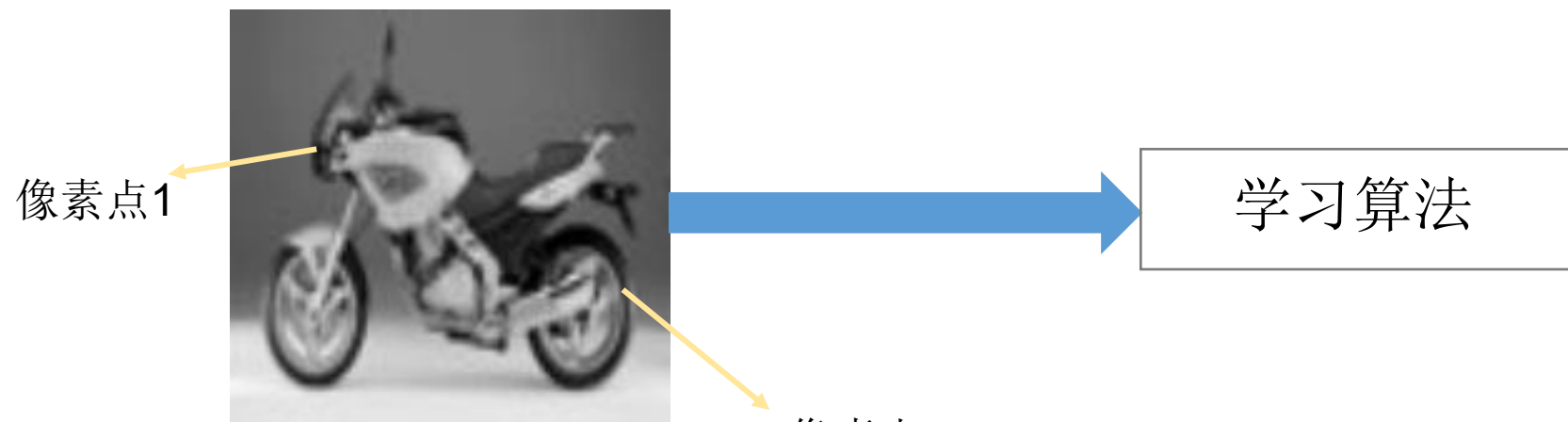
数据与特征



数据与特征



数据与特征

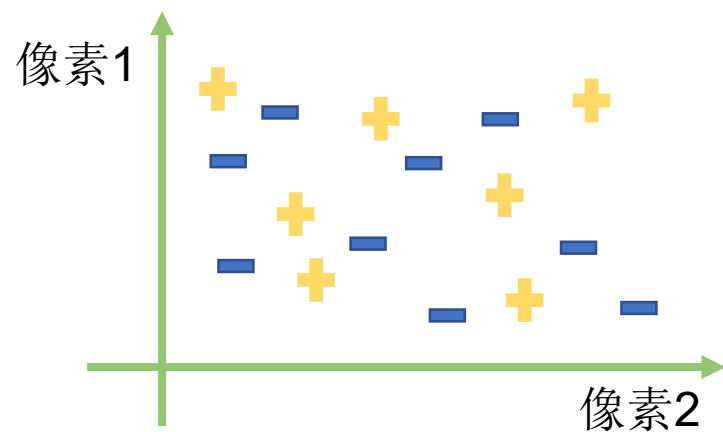


输入

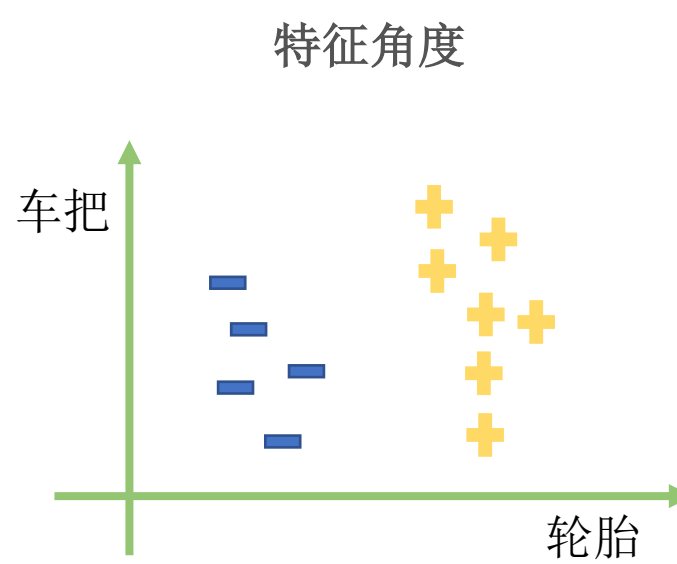
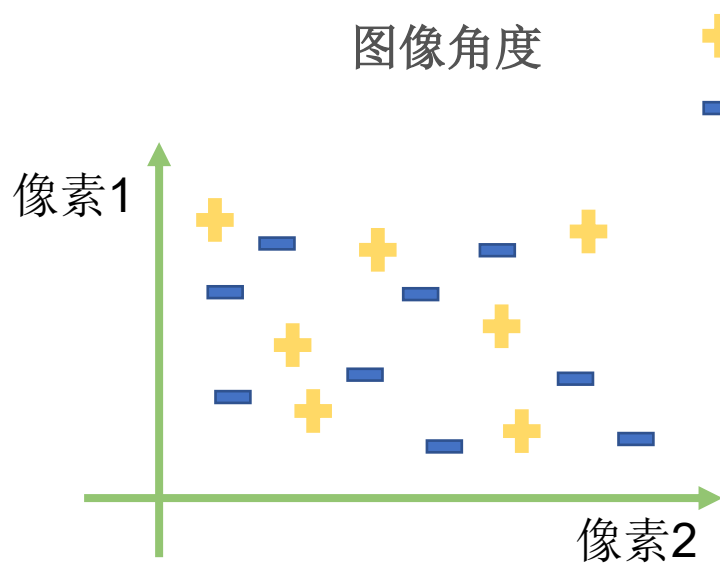
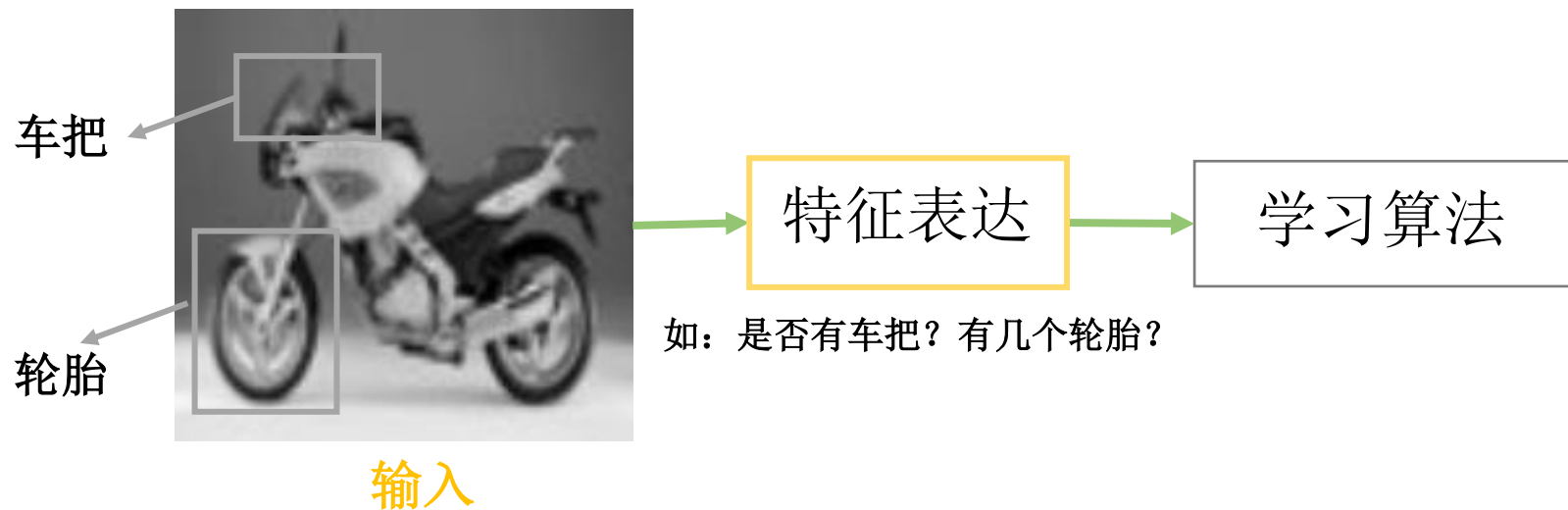
像素点2

图像角度

+ 摩托车
— 非摩托车

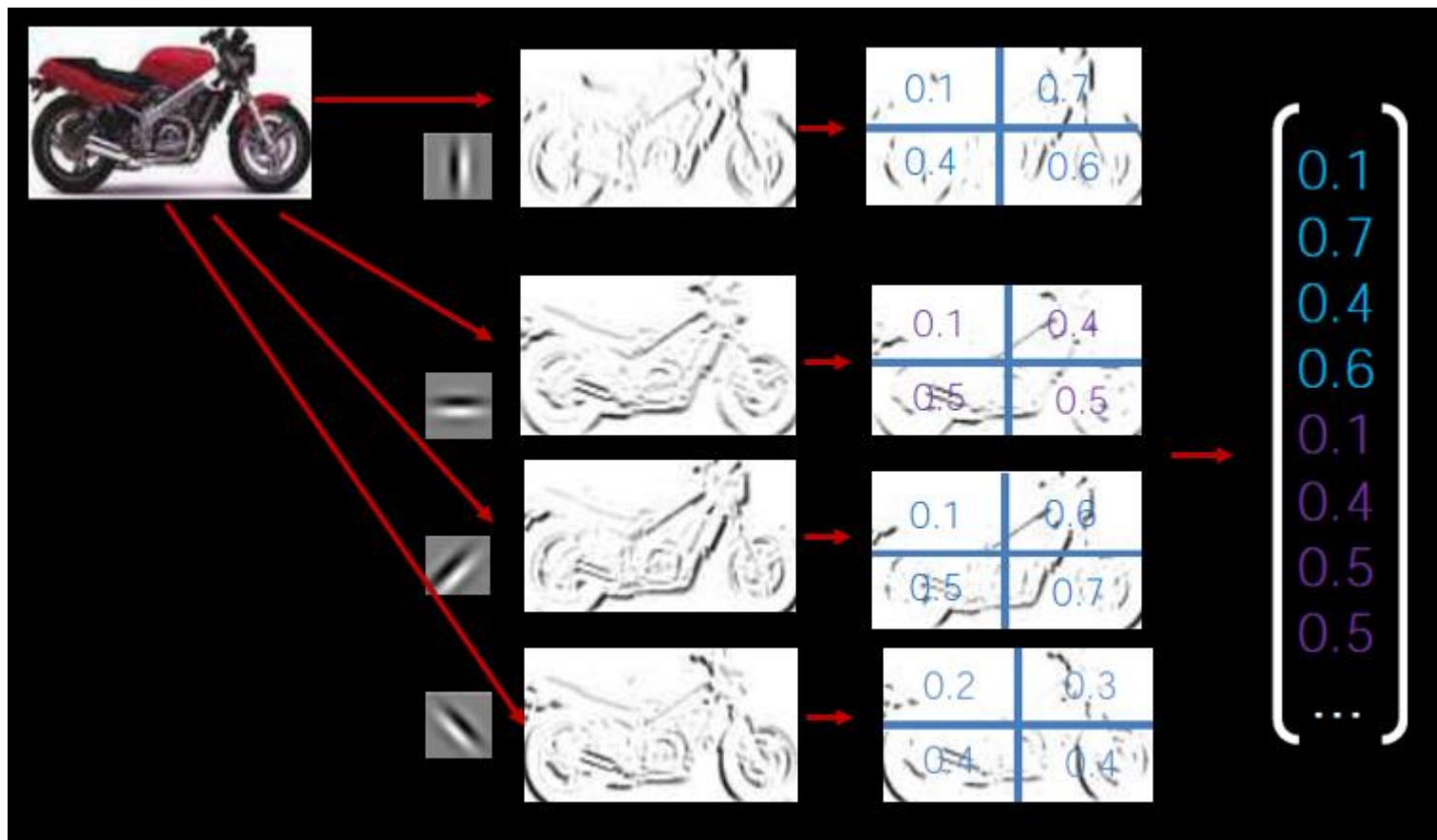


特征表达



特征表达

然而，我们并没有“车把”检测器。所以很早的机器学习领域专家通过“手工”（hand-design）设计特征来捕捉图片的统计学特征。



能做什么？

图片/视频



图像



图像特征

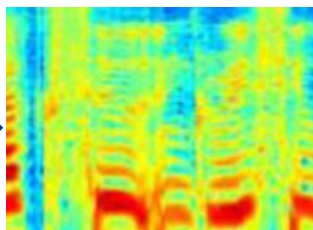


检测

音频



音频



声音特征

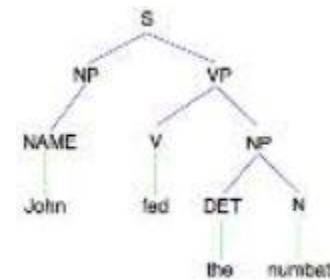


发声人

文字



文字



文字特征

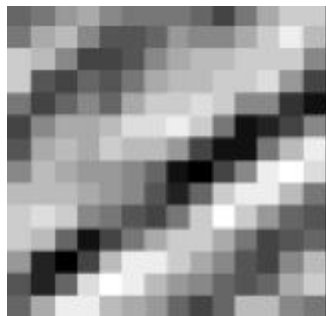


文字分类,
机器翻译,
信息检索

.....

小问题

我们有一个 14×14 像素大小的图像，可以把这个图像用196个数字来表示



$\begin{bmatrix} 255 \\ 98 \\ 93 \\ 87 \\ 89 \\ 91 \\ 48 \\ \dots \end{bmatrix}$

我们可不可以学到更好的特征向量来表达它呢？

学习算法



输入

特征表达

学习算法

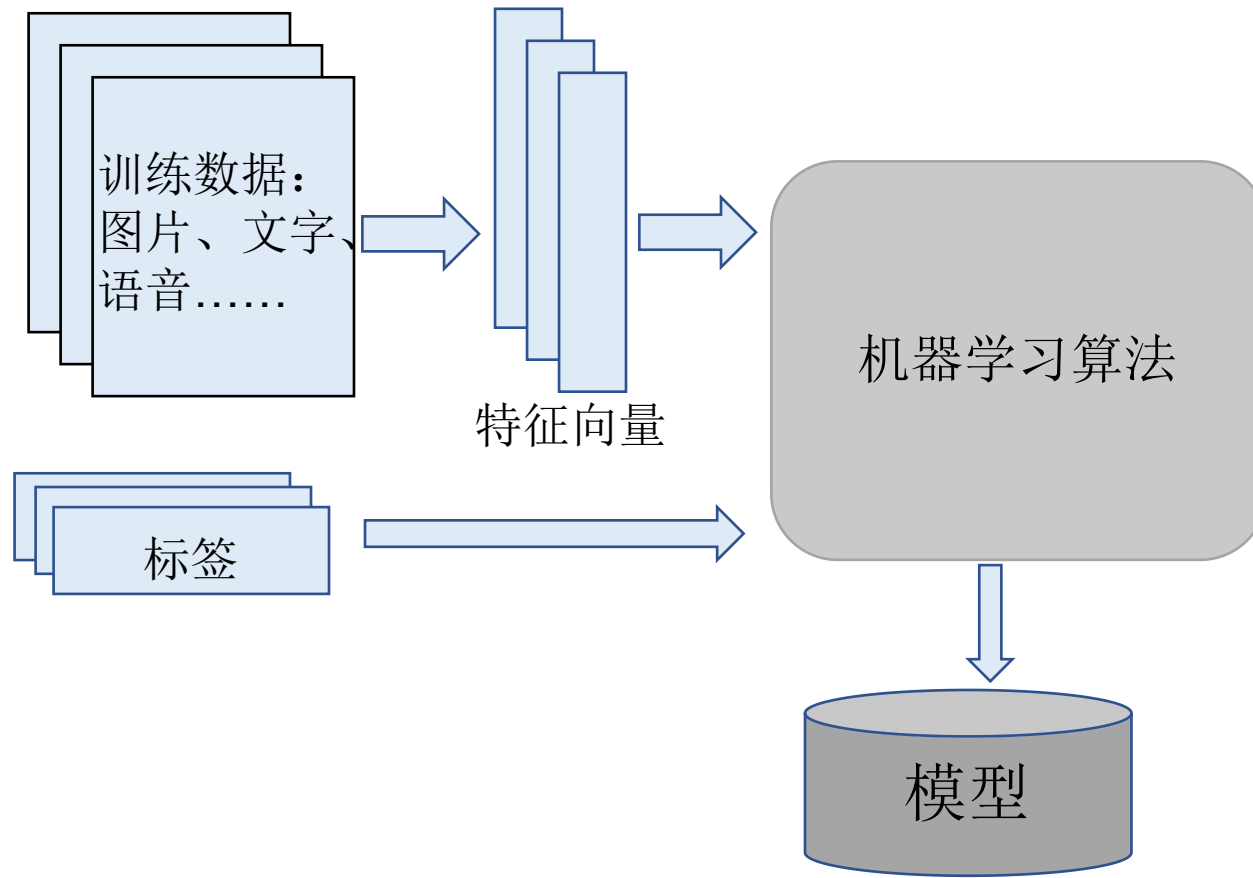
$\begin{bmatrix} 255 \\ 98 \\ 93 \\ 87 \\ 89 \\ 91 \\ 48 \\ \dots \end{bmatrix}$

逻辑回归、
支持向量机
等算法

最终模型

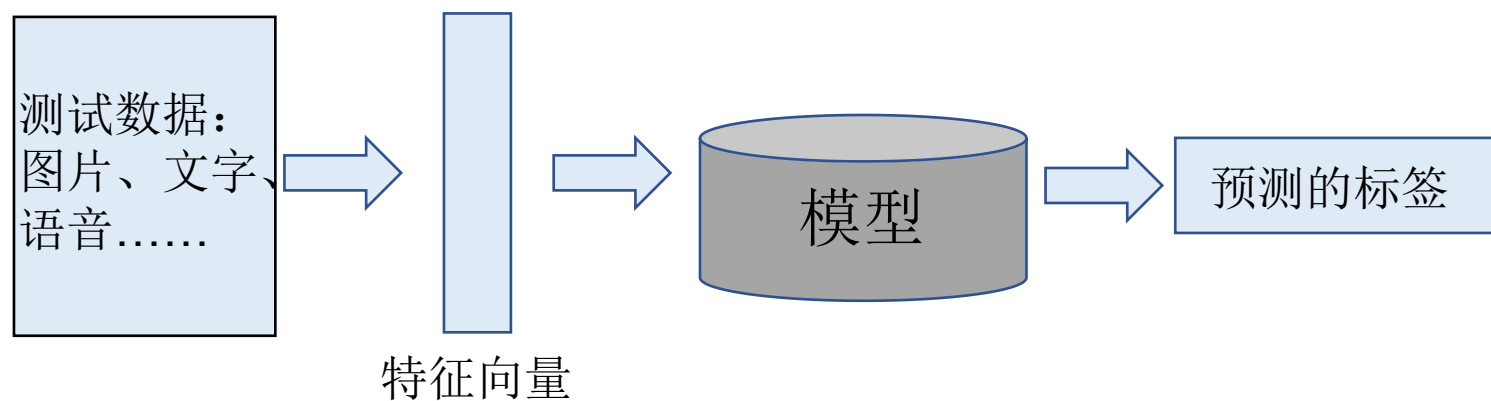
典型的机器学习过程

建模：

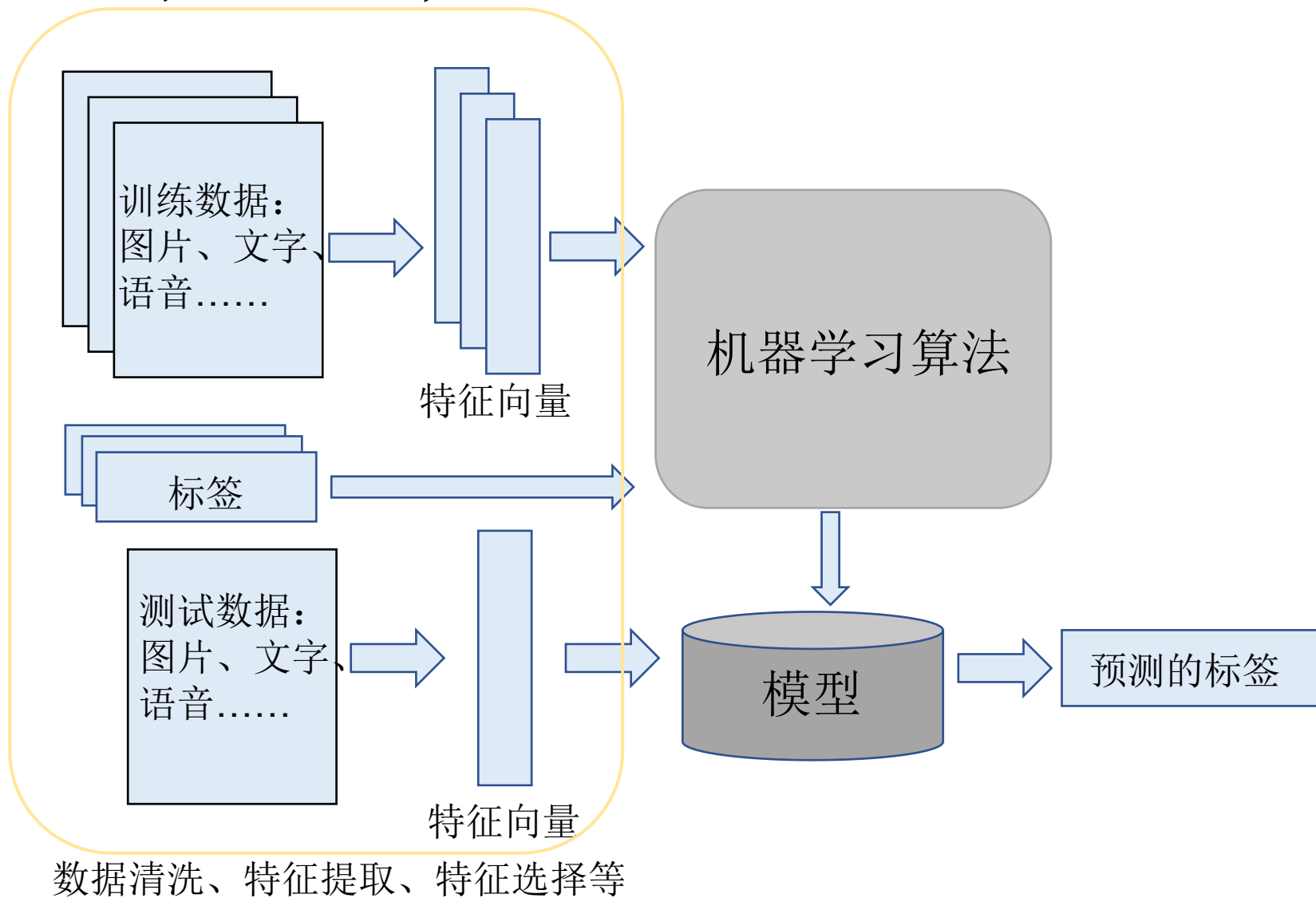


典型的机器学习过程

预测：



典型的机器学习过程



典型的机器学习过程

